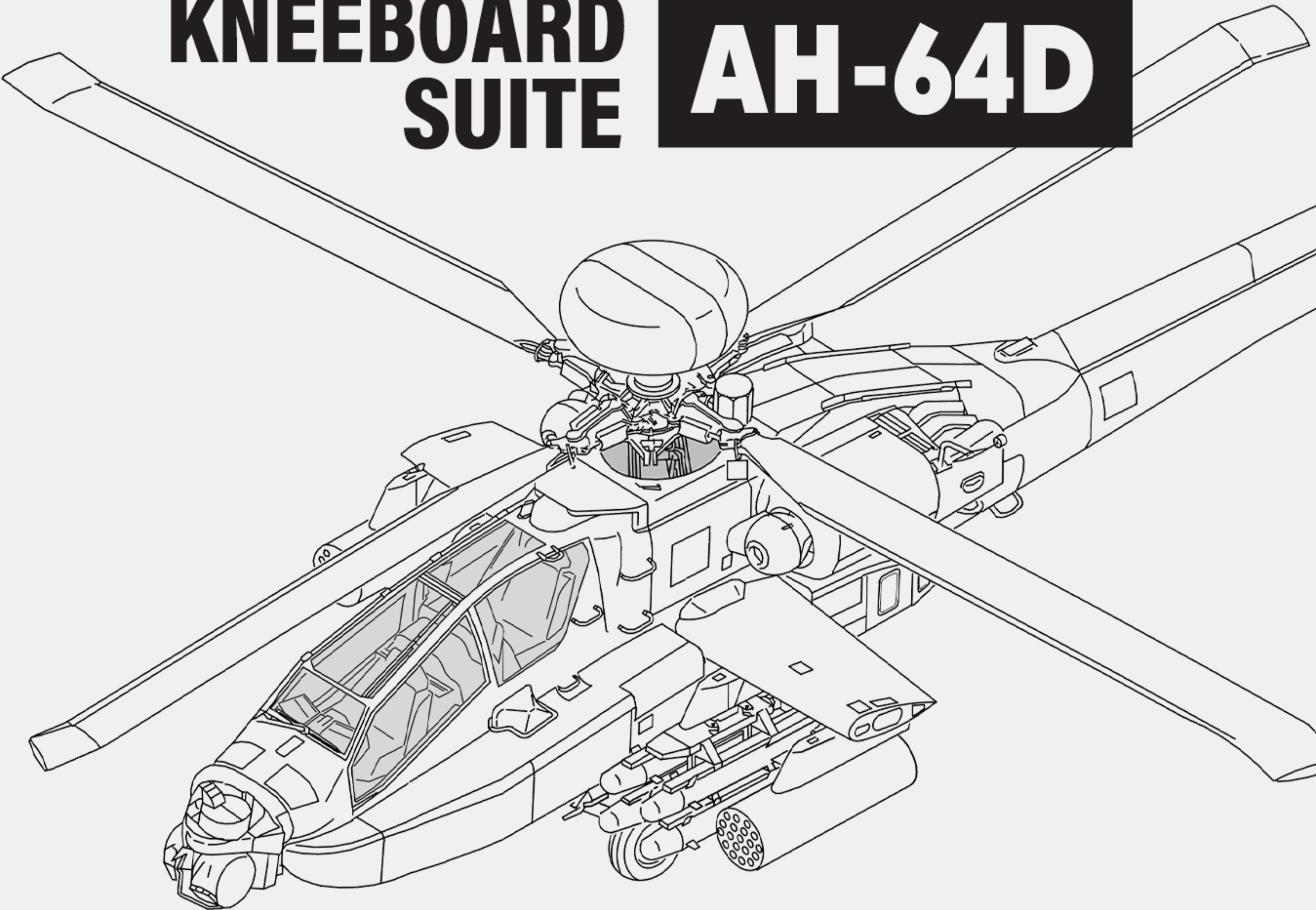


KNEEBOARD SUITE

AH-64D



CHECKLISTS, Pt. 1. DMS SWEEP. 2

CHECKLISTS, Pt. 2. WAILRM. COORDS. 3

ASE THREATS 4

GEORGE AS CPG 5

GEORGE AS PILOT 6

ORDNANCE. REARMING. 7

REFUELING. FUEL TRANSFER. 8

TSD SYMBOLS, Pt. 1 9

TSD SYMBOLS, Pt. 2. FCR TARGETS. 10

CPG STATION. RADIOS. 11

PILOT STATION. QUICKSTART. 12

PLT	INTERIOR CHECK
1	EXT / INTR lights, COMM panel, KU, MPD, and EUFD knobs As desired
2	PARK BRAKE <i>handle out</i> Set
3	POWER levers, RTR BRK switch, NVS MODE switch OFF

CPG	INTERIOR CHECK
1	INTR lights, COMM panel, TEDAC, KU, MPD, and EUFD knobs As desired
2	POWER levers, NVS MODE switch OFF

ONCE BATTERY POWER IS AVAILABLE:
SIGNAL LIGHTS TEST. FIRE TESTS 1 & 2.

PLT	STARTING APU
BATTERY TIME UNDER LOAD IS LIMITED BY FAT TO: >43°C = 1 MIN 32-43°C = 3 MIN <32°C = 20 MIN	
1	MSTR IGN switch <i>rmb</i> BATT
2	TAIL WHEEL Verify locked

SIGNAL LIGHTS TEST. FIRE TESTS 1 & 2.

3	APU push button Press <i>'APU ON' EUFD advisory will appear within 20 sec</i>
---	--

PLT	AFTER STARTING APU
1	Canopy door Closed
2	RLWR volume As desired
3	SQL momentary switches <i>rmb</i> Flip forward
4	DMS > DTU page > Select MASTER LOAD
5	DMS sweep Perform
6	CMWS Control Panel As desired
7	NVS MODE switch As desired
8	SAI <i>scroll down</i> Uncaged

CPG	AFTER STARTING APU
1	Canopy door Closed
2	DMS sweep Perform

- IHADSS BORESIGHTING
- 1 [SIGHT SELECT SWITCH] — to HMD

2 WPN > BORESIGHT page > IHADSS

3 INTR LT > Adjust PRIMARY knob
so that the symbol on BRU is visible

4 Align HMD reticle with BRU

5 Press B/S NOW to confirm position



THE SWEEP ITEMS ARE SPLIT AS BRIEFED. TYPICALLY, THE PILOT TAKES COMMS AND NAVIGATION, AND THE CPG TAKES WEAPONS AND SENSORS. THE SWEEP MAY BE FINISHED AFTER TAKEOFF.

DMS SWEEP	
ASE > UTIL	WPN > MANRNG
▶ RLWR — ON	▶ '800', 'A', or as desired
▶ Choose CHAFF MODE and adjust PROGRAM	WPN > GUN
▶ Adjust flare program (via the kneeboard)	▶ Select BURST LIMIT
TSD > SHOW	WPN > MSL
▪ Configure SHOW, THRT SHOW, and COORD SHOW for NAV and ATK phases	▶ If self-lasing is desired: PRI code = LRFD code
TSD > UTIL	WPN > RKT
▪ TIME — ZULU or LOCAL	▶ Select INVENTORY type and QTY
▪ Verify SYSTEM TIME	WPN > BORESIGHT
▪ DOPPLER — verify ON	▶ IHADSS < PILOT < CPG
TSD > POINT	FCR > UTIL
▪ Check/add points and targets	▶ MMA (if avail.) — NORM
TSD > RTE	
▪ Check/adjust route, set DIR as needed	FLT > SET
TSD > BAM	▶ RDR ALT — verify ON
▪ Set up Priority Fire and No-Fire zones	▶ Set HI/LO alt alerts
TSD > INST > UTIL	▶ Set ALT to field elevation or set PRESS to QNH
▶ ADF — ON	▪ Set standby altimeter
	▪ UNITS — as desired
	▪ Horizon line — as desired
	FUEL
	▪ AUX GALLONS EXT: enter the total amount of fuel in all external tanks (230, 460, 690, 920)
	UTIL
	▪ After closing both doors: ECS — verify ON, temp set as desired (50–90°F)
	COM
	▶ Radios — check/tune
	▪ UHF GUARD REC — ON
	▪ XPNDR — as desired <i>MP</i>
	▪ Verify ORIG ID and CALL SIGN
	▪ Set up datalink preset: RETRIES 0, TUNE SC, select on the EUFD, etc.
	▪ Set up team members
	▪ Test datalink: TSD > RPT > PP > MSG RQST > SEND
WPN > CODE > FREQ	
▪ Modify laser codes as needed. For example: A — freq for self-lasing, B — wingman's freq, C — JTAC's freq.	
WPN > CODE	
▪ Assign laser codes as needed. For example: LRFD — A, LST — B or C.	
WPN > CHAN	
▪ Assign laser codes to the missile channels as needed, or leave as is: 1 — A, 2 — B, 3 — C.	
WPN > UTIL < PILOT	
▶ IHADSS, PNVS — ON	
WPN > UTIL < CPG	
▶ IHADSS, TADS, FLIR, and LASER — ON	

PLT	STARTING ENGINES
1	Control sweep, trim check Perform
2	ANTI-COL lights WHT (<i>day</i>), RED (<i>night</i>) or OFF (<i>when required</i>)
3	First Engine START switch <i>rmb</i> START
4	At first indication of Ng increase: POWER lever IDLE
5	Once first engine is stabilized at 66–67% Ng: Second Engine START switch <i>rmb</i> START
6	At first indication of Ng increase: POWER lever IDLE
7	Once both engines are stablilized: POWER levers Advance smootly to FLY
8	Once Np and NR are stabilized at 101%: APU Off
BEFORE TAXI CHECK	
1	TSD page ENSURE THE MAP IS DISPLAYED (EGI "ALIGNMENT" TAKES 5 MIN WITH OR WITHOUT GPS)
2	Chocks Removed
3	ELITE ON / As required
4	PARK BRAKE Release
5	TAIL WHEEL Unlock
6	Wheel brakes (<i>both seats</i>) Check during taxi
BEFORE TAKEOFF CHECK	
1	Weapons Not actioned, SAFED, ORIDE off
2	TAIL WHEEL button Locked, no light
3	PARK BRAKE As desired
4	POWER levers To FLY
5	XPNDR As desired
6	BINGO or ADF timer As desired
7	Power check Perform (<i>validate PERF page</i>)
LEVEL OFF CHECK	
1	WAILRM–NCA Perform
2	FUEL page > AUX transfer As desired
3	DMS sweep (<i>if not already done</i>) Finish
BEFORE LANDING CHECK	
1	Weapons Not actioned, SAFED, ORIDE off
2	ASE & CMWS Control Panel As required
3	TAIL WHEEL Locked
4	PARK BRAKE Released

PLT	ENGINE SHUTDOWN (APU)
IF ENGINES ARE NOT IDLED FOR 2 MINUTES PRIOR TO SHUTDOWN, A RESTART SHOULD BE AVOIDED BETWEEN 5 MINUTES AND 4 HOURS.	
1	APU button Press ON
2	TAIL WHEEL / PARK BRAKE Lock / Set
3	After 'APU ON' EUFD advisory is displayed: POWER levers IDLE
4	SAI <i>lmb</i> + scroll up Cage
5	NVS MODE switch OFF
6	ACM switch VIDEO panel OFF
7	WPN > UTIL page > PNVS OFF
8	After idling for 2 minutes: POWER levers OFF
9	Once below 50% NR: RTR BRK switch BRK
10	Stabilator (ENG > SYS > STAB) Set to ZERO
11	When rotor stops: RTR BRK switch OFF
12	EXT / INTR lights, Searchlight OFF
13	APU button OFF
14	MSTR IGN switch OFF
15	Before leaving the helicopter Set chocks
CPG	ENGINE SHUTDOWN (APU)
1	NVS MODE switch OFF
2	TEDAC display knob OFF
3	DMS > SHUTDOWN page > MASTER OFF
4	INTR lights OFF

(W) WEAPONS
(A) ASE

- CMWS
- CHAFF
- RLWR

(I) IFF
(L) LIGHTS
(R) RECORDER
(M) MPDs, ACQ

(N) NAV
(C) COMMS
(A) AIR SURV

- C-SCOPE

UTM > 4+4 MGRS
TSD DISPLAY FORMAT:
47 00A BC 1234 5678
FOUR-DIGIT EASTING & NORTHING
KU ENTRY FORMAT:
00ABC12345678
TO CHANGE DATUM (47): TSD > UTIL

LAT/LONG > DD°MM.mm
DCS F10 MAP FORMAT:
N12°34.567 E12°34.567
LLDM (LAT LONG DECIMAL MINUTES)
TSD DISPLAY FORMAT:
N12 34.56 E012 34.56
DEGREES OF LONGITUDE IS ALWAYS THREE-DIGIT (ADD LEADING ZERO).
DECIMALS ARE ALWAYS TWO-DIGIT.
KU ENTRY FORMAT:
N123456E0123456

SURFACE		NAVAL	MISSILES		nm / km		AIRBORNE		
2	SA-2 Guideline S-75	Amazonka RF	/ Fan Song TR	C	19/35	14	F-14A / GR/B	123	
3	SA-3 Goa S-125		Low Blow TR	C	08/15	15	F-15C	123	
5	SA-5 Gammon S-200		Square Pair TR	C	135/250		F-15E	23	
6	SA-6 Gainful Kub		Straight Flush STR	C	16/30		F-15E S4+	123	
7	HQ-7B		SR / TELAR	E	07/13	16	F-16A	123	
8	SA-8 Gecko Osa		TELAR	C	06/11		F-16A MLU	23	
10	SA-10 Grumble S-300PS:						F-16C / CM	23	
	Big Bird SR	Clam Shell SR	/ Flap Lid A & B TR	C	25/46	18	F/A-18A	12	
11	SA-11 Gadfly Buk	Snow Drift SR	/ Fire Dome TELAR	C	16/30		F/A-18C	123	
13	SA-13 Gopher Strela-10M		TELAR	F	02/04	19	MiG-19P	2	
15	SA-15 Gauntlet Tor		TELAR	C	07/13	21	MiG-21Bis	12	
	SA-15 Gauntlet M2 Tor-M2		TELAR	C	08/15	23	MiG-23MLD	12	
22	SA-22 Greyhound Pantsir-S1		SPAAW	C	11/20	25	MiG-25PD	12	
AA	Fire Can SON-9	AAA Fire Director Radar				29	MiG-29A/G	12	
GS	Gepard		SPAAA	E	03/06		MiG-29S	123	
GS	LPWS Centurion Phalanx		C-RAM	E	02/04		Su-27/33	12	
GS	Vulcan		SPAAA	E	02/04		J-11A	123	
HK	Hawk	CWAR	/ SR / TR	C	11/20	30	Su-30	123	
IT	IRIS-T SLM		STR	F	22/41	31	MiG-31	12	
MM	MISSILE					34	Su-34	123	
NV	NAVAL UNIT					37	AJS-37 Viggen	2	
PT	Patriot		STR	C	23/42	50	A-50 / KJ-2000	AWACS	
RA	Rapier with Blindfire Radar		TR	E	05/10	E2	E-2D	AWACS	
	Rapier with or without optical Tracker		LN	E	05/10	E3	E-3A	AWACS	
RO	Roland		EWR	/ ADS	C	06/11	F1	Mirage F1	12
S6	SA-19 Grison Tunguska		SPAAW	E	05/10	F2	Tornado GR4	2	
SA	NASAMS with AIM-120B		SR	C	07/13	F4	F-4E-45MC	12	
	NASAMS with AIM-120C		SR	C	08/15	F5	F-5E / E-3 / FC	2	
SR	Nebo 1L13	Nebo 55G6	/ AN/FPS-117	EWR		JF	JF-17	23	
TR	Dog Ear Sborka	[13-15-19]	MCC-SR			M2	M-2000C	12	
	Flat Face B	[2-3-5]	SR				Mirage 2000-5	123	
	Tin Shield	[5-10]	SR			U	A-6E	2	
U	HQ-7 'Player' AI- or user-controlled		TELAR	E	07/13		AH-64D (old)	3	
	Freya	Würzburg	WWII Assets Pack	EWR			C-101CC	2	
	Silkworm		anti-ship	SR			H-6J	BOMBER	
	Tall King	[5]	SR				Tornado IDS	2	
ZU	ZSU-23-4 Shilka		SPAAA	E	02/04		Tu-95MS	BOMBER	
							SH-60B	HELO	

1 — SARH, 2 — IR, 3 — ARH — SUPPORTED A-A MISSILES. CM TO USE: C — CHAFF, F — FLARES, E — EVADE.
19/35 — SAM'S MAX FIRING RANGE vs. AH-64D AT 3000 ft, nm / km.
[THE SAM] TO WHICH THIS RADAR IS USUALLY LINKED.
IN GREEN — SUPPORTING UNIT: DESTROYING IT MIGHT NOT DISABLE THE SITE.

XX

SURFACE THREAT

XX

AIRBORNE THREAT

XX

ACQUISITION: BOXED ICON

XX

TRACKING: BOXED + DASHED LINE

XX

LAUNCH: FLASHING BOX

*

LWR THREAT

*

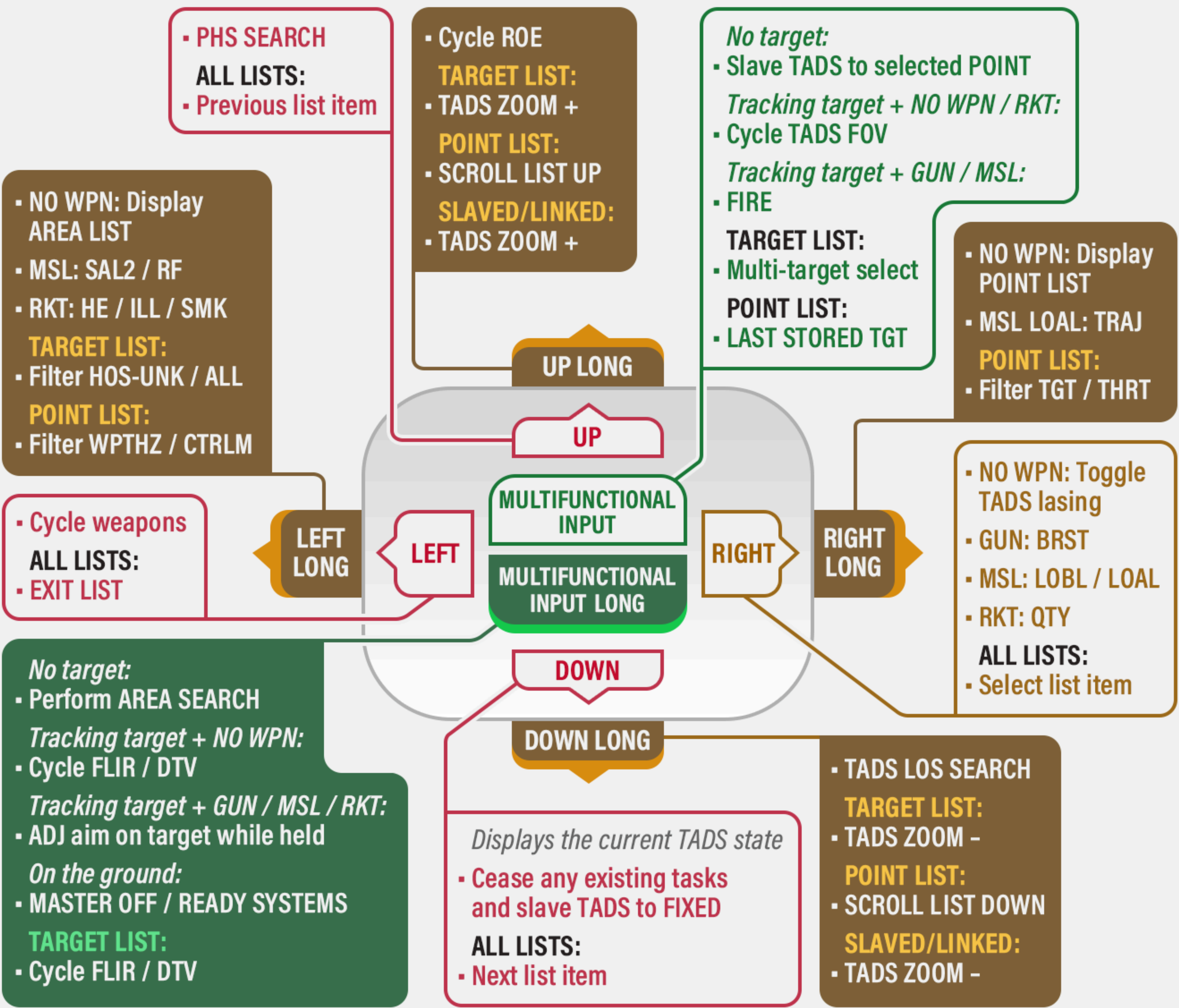
RANGING: BOXED ICON

*

DESIGNATING: BOXED + DASHED LINE

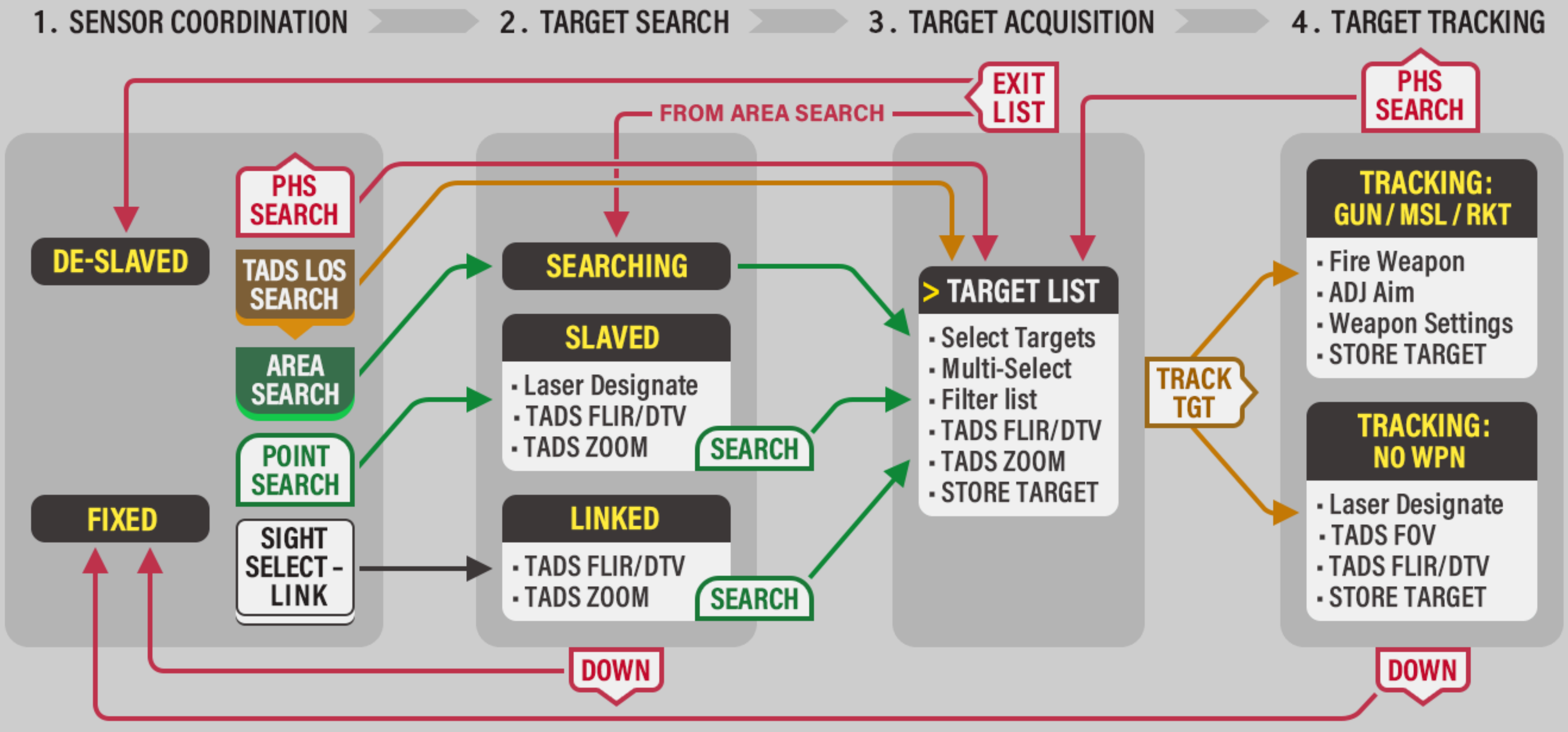
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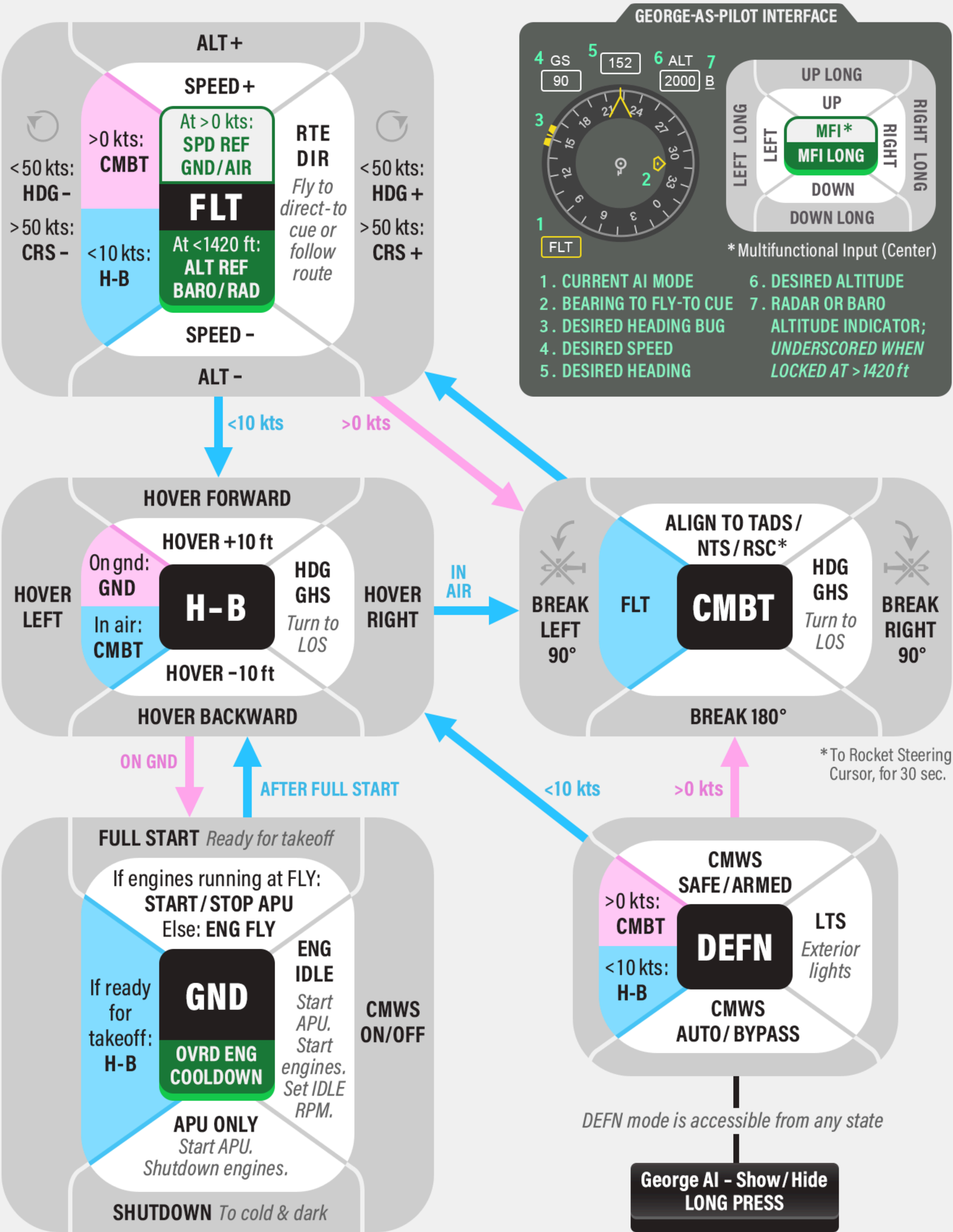
BEAMING: FLASHING BOX



NORMAL ACTIONS (IN BROWN) AND COORDINATED ACTIONS (IN GREEN) ARE BROADLY CONTEXTUAL DEPENDING ON THE CURRENT TASK. PRIORITY ACTIONS (IN RED) ARE CONSTANT.

GEORGE-AS-CPG WORKFLOW





RAPID REARMING

- 1 Prior to entering the FARP:
Weapons **SAFED, ORIDE OFF**
- 2 WPN > UTIL page:
MSL, RKT, GUN **OFF**
- 3 TAIL WHEEL **Locked**
- 4 PARK BRAKE **Set**
- 5 IDM and HF:
Auto transmit **Disable**
- 6 WPN > UTIL page:
MSL, RKT, GUN **ON**
- 7 FUEL page: **230 / 460 / 690 / 920**
AUX GALLONS EXT **Update**
- Rearming completed:

NM = KM ÷ 1.85

M = FT ÷ 3.28

KM = NM × 1.85

FT = M × 3.28

1 KM = 0.54 NM

LBS = GAL × 6.66

GROUND SPEED REQUIRED (KTS) =
(DISTANCE ÷ MINUTES) × 60

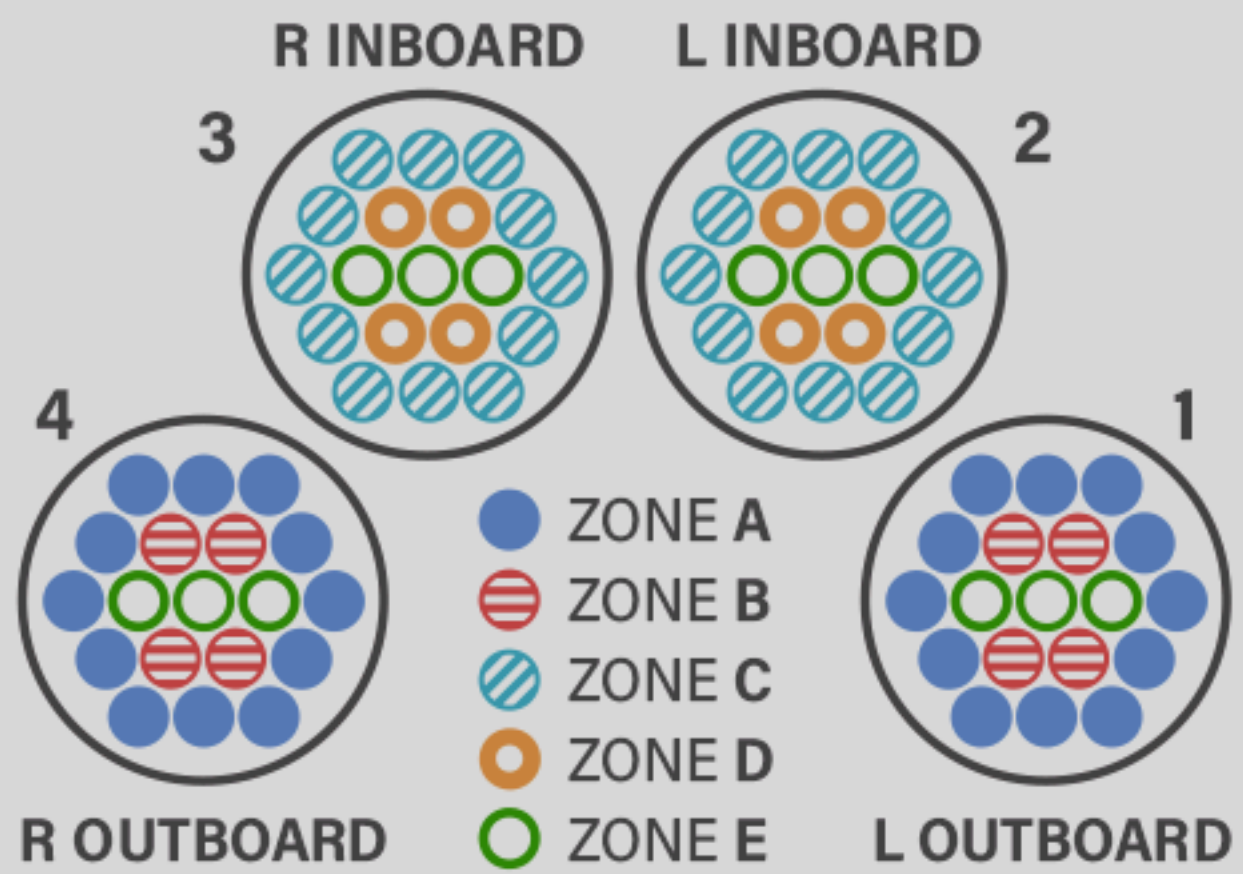
TIME OF FLIGHT (MINS) =
(DISTANCE ÷ GROUND SPEED) × 60

BINGO FUEL (LBS) =
(TIME OF FLIGHT ÷ 60) × FUEL LB/HR

OBJECTIVE TIME (MINS) =
([TOTAL FUEL - BINGO FUEL] ÷ FUEL LB/HR) × 60

SFR (SPECIFIC FUEL RANGE) FACTOR =
GROUND SPEED ÷ FUEL LB/HR

FLIGHT RANGE (NM) =
SFR × TOTAL FUEL



EMPTY WEIGHT (NO MMA)	13074
MAX WEIGHT	23000
MAX FUEL CARRIED:	
FWD INTERNAL TANK	1040
AFT INTERNAL TANK	1460
IAFS (ROBBIE TANK)	650
EXTERNAL TANKS	4x 1525
	9250

U.S. Army AH-64D Block 2 (2005-2010)
Lot 9.1 avionics, T-700-GE-701C engines

524 Mast Mounted Assembly
FCR/RFI

0 30 chaff + 60 flares
non-editable

980 98 GAL IAFS (ROBBIE) C AUX
Limits gun ammo to 300 rounds

M230E1 30mm gun

• M789 HEDP high explosive

• M788 TP target practice

227 Up to 300 rounds (with IAFS)

908 Up to 1200 rounds (w/o IAFS)

A/G MISSILES

144 M299 Hellfire Launcher

100 AGM-114K Hellfire SAL

108 AGM-114L Longbow Hellfire RF

M261 ROCKET PODS

SINGLE TYPE HYDRA 70/Mk 66

523 19x M151 HE 6PD
10 lb warhead, soft targets

523 19x M151 HE, M433 RC 6RC
custom delay, soft targets

668 19x M229 HE 6PD
17 lb warhead, soft targets

556 19x M257 IL 6IL
illumination flare (3 min),
deployed 3500 m
from the launch point

523 19x M274 TP-SM 6SK
target practice

607 19x M282 MPP 6RC
penetrator, light armor

MIXED LOAD, INBOARD

527 C: 12x M257 IL 6IL
D/E: 7x M151 6PD

522 C: 12x M274 TP-SM 6SK
D/E: 7x M151 6PD

MIXED LOAD, OUTBOARD

527 A/B: 16x M151 6PD
E: 3x M257 IL 6IL

522 A/B: 16x M151 6PD
E: 3x M274 TP-SM 6SK

FUEL TANKS

1690 230 GAL EXT TANK L AUX R AUX
165 lbs empty tank,
1525 lbs of fuel,
no ballistic protection,
no quantity probes

NOTE: WITHOUT THE INBOARD
TANK INSTALLED, THE FUEL IN
ITS ADJACENT OUTBOARD
TANK WILL BE INACCESSIBLE.

RAPID (HOT) REFUELING

ENGINE 2 SHUT DOWN, OR BOTH OPERATING

- 1 Prior to entering the FARP:
Weapons SAFED, GND ORIDE off
- 2 TAIL WHEEL / PARK BRAKE Locked / Set
- 3 POWER levers:
A Both engines running:
1) POWER lever **NO 2** IDLE
2) FUEL page > XFER OFF
B Single engine running:
1) POWER lever **NO 2** IDLE for 2 minutes, then OFF
- 4 IDM and HF auto transmit Disable
- 5 ANTI-COL switch OFF
- Refueling completed:
- 6 IDM and HF auto transmit As required
- 7 ANTI-COL switch As desired
- 8 FUEL page > AUX GALLONS EXT Update

NON-SANDY ENVIRONMENT

- 9 (Bubble Burn) Apply collective until 60% torque is reached or aircraft is light on wheels. Maintain power settings for 60 sec.
- 10 Collective Reduce to minimum torque
- 11 POWER levers:
A Both engines running:
1) POWER lever **NO 2** To FLY
2) FUEL page > XFER AUTO
B Single engine running (cross-bleed start):
1) Collective Increase to attain 95% NG2
2) **Engine 2** Start
3) Collective Reduce to flat pitch after starter dropout
4) POWER lever **NO 2** To FLY

RAPID (WARM) REFUELING

BOTH ENGINES SHUT DOWN AND APU ON

- 1 Prior to entering the FARP:
Weapons SAFED, GND ORIDE off
- 2 TAIL WHEEL / PARK BRAKE Locked / Set
- 3 APU Start (wait for 'APU ON' on EUFD)
- 4 POWER levers IDLE for 2 minutes, then OFF
- 5 IDM and HF auto transmit Disable
- 6 Once below 50% NR:
RTR BRK switch BRK until rotor stops, then OFF
- 7 ANTI-COL switch OFF
- Refueling completed:
- 8 IDM and HF auto transmit As required
- 9 FUEL page > AUX GALLONS EXT Update
- 10 Perform STARTING ENGINES procedure **Pt.2**

SANDY ENVIRONMENT

- 9 POWER levers:
A Both engines running:
1) POWER lever **NO 2** To FLY
2) FUEL page > XFER AUTO
B Single engine running:
1) APU Start
2) **Engine 2** Start
3) POWER lever **NO 2** To FLY
- 10 FUEL page > CROSSFEED Switch:
1) To FWD, wait 60 sec.
2) To NORM, wait 30 sec.
3) To AFT, wait 60 sec
4) To NORM, wait 30 sec.

FUEL TRANSFER

WHEN TO BEGIN FUEL TRANSFER

FROM C AUX:

Typically, after the second 30-minutes BINGO alert (FUEL > CHECK), or after the TOT drops below 2500 lbs.

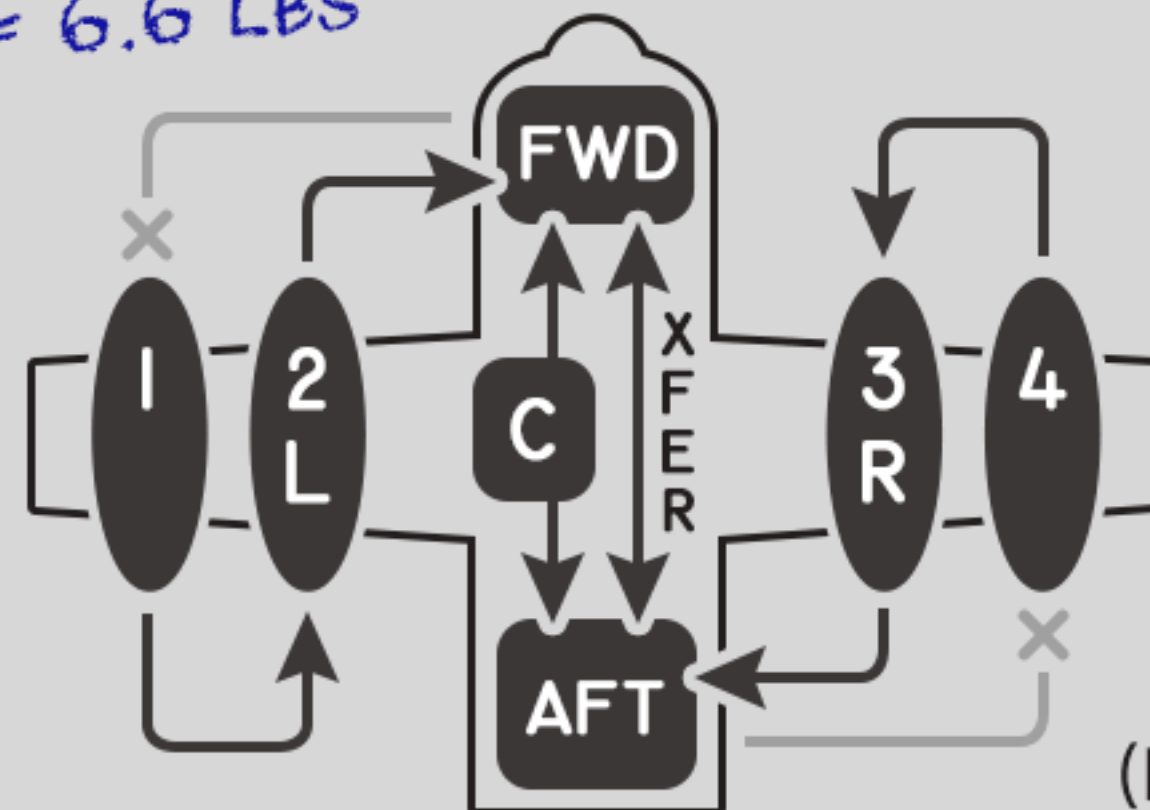
FROM EXT AUX:

Typically, after takeoff.

FUEL TRANSFER PRIORITY:

- 1. FWD / AFT (XFER)
 - 2. EXT > FWD / AFT
 - 3. C AUX > FWD / AFT
- Meaning, transfer from the C AUX will not start until the FWD / AFT / EXT transfer is completed or suspended.

1 GAL = 6.6 LBS



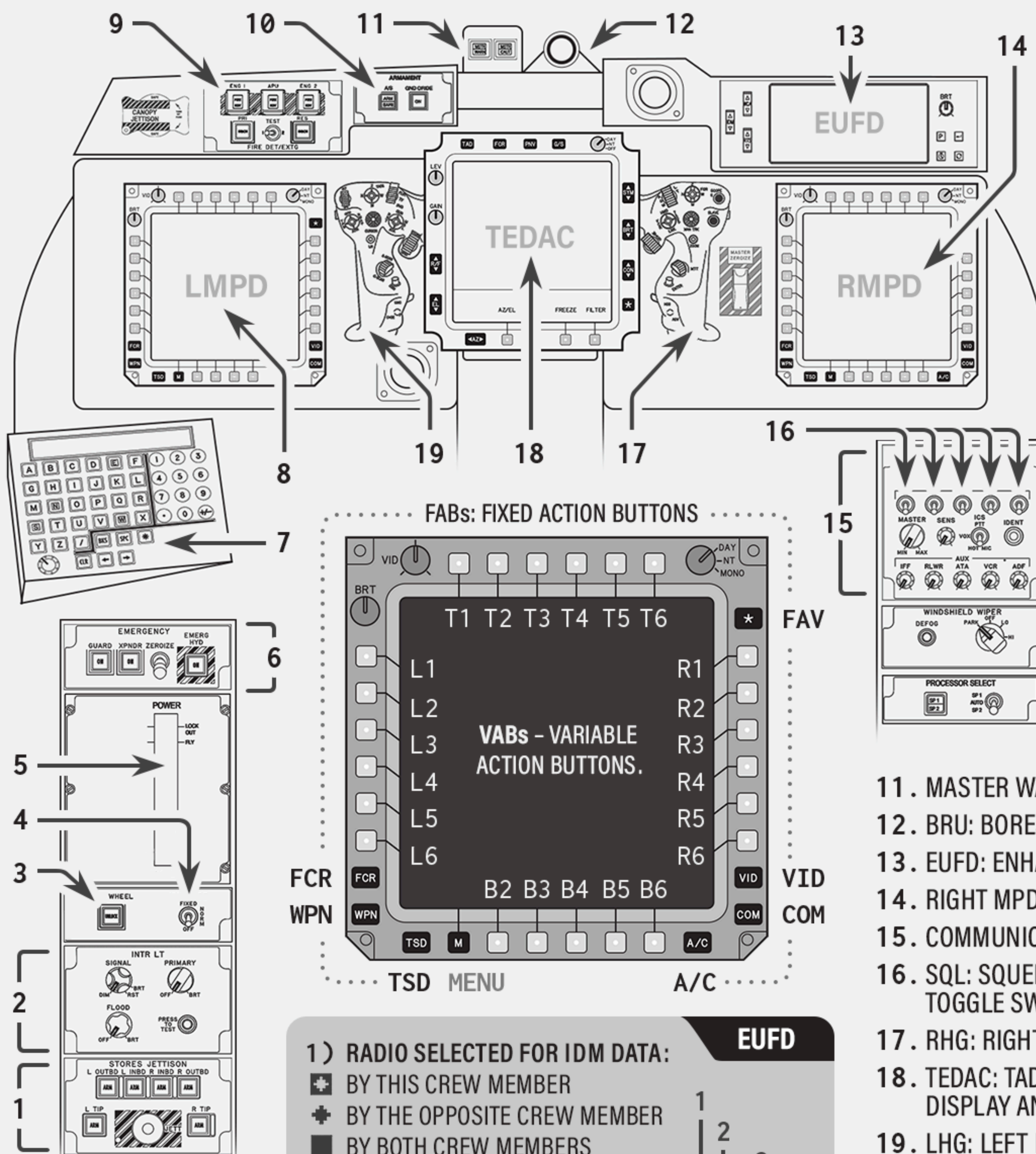
FWD	1040 lbs
	156 gal
AFT	1460 lbs
	220 gal
C AUX	650 lbs
(ROBBIE)	98 gal
EXT AUX	1525 lbs
(EACH TANK)	230 gal

NOTE: THE **OUTBOARD** EXT TANKS **1** AND **4** CAN TRANSFER FUEL **ONLY** INTO THEIR RESPECTIVE **INBOARD** EXT TANKS, WHICH THEN FEED THE INTERNAL FUEL TANKS. MEANING, **1** WITHOUT **2** AND **4** WITHOUT **3** ARE DEAD WEIGHT.

WP HZ											
WAYPOINTS, HAZARDS		01-50									
				ASSEMBLY AREA	AA		FRIENDLY AIR DEFENSE	<u>AD</u>		ENEMY AIR DEFENSE	<u>ED</u>
	COMMS CHECK POINT	CC		BATTALION	BN		FRIENDLY AIR ASSAULT	<u>AS</u>		ENEMY AIR ASSAULT	<u>ES</u>
	LANDING ZONE	LZ		BATTLE POSITION	BP		FRIENDLY AIR CAVALRY	<u>AV</u>		ENEMY AIR CAVALRY	<u>EV</u>
	PASSAGE POINT	PP		BRIDGE OR GAP	BR		FRIENDLY AIRBORNE	<u>AB</u>		ENEMY AIRBORNE	<u>EB</u>
	RELEASE POINT	RP		BRIGADE	BD		FRIENDLY ARMOR	<u>AM</u>		ENEMY ARMOR	<u>AE</u>
	START POINT	SP		CHECKPOINT	*CP		FRIENDLY AR-MORED CAVALRY	<u>CA</u>		ENEMY AR-MORED CAVALRY	<u>EC</u>
	WAYPOINT	*WP		COMPANY	CO		FRIENDLY AVIATION MAINT.	<u>MA</u>		ENEMY AVIATION MAINT.	<u>ME</u>
	TOWER OVER 1000' AGL	TO		CORPS	CR		FRIENDLY CHEMICAL	<u>CF</u>		ENEMY CHEMICAL	<u>CE</u>
	TOWER UNDER 1000' AGL	*TU		DIVISION	DI		FRIENDLY DE-CONTAMINATION	<u>DF</u>		ENEMY DE-CONTAMINATION	<u>DE</u>
	WIRES: POWER TRANSM. LINES	WL		FARP: FUEL ONLY	<u>FF</u>		FRIENDLY ENGINEERS	<u>EN</u>		ENEMY ENGINEERS	<u>EE</u>
	WIRES: PHONE AND ELECTRIC	WS		FARP: AMMO ONLY	<u>FM</u>		FRIENDLY ELECT-RONIC WARFARE	<u>FW</u>		ENEMY ELECT-RONIC WARFARE	<u>WR</u>
*CURSOR DROP DEFAULTS TO THIS SYMBOL UNDERLINED ITEMS ARE ABLE TO DISPLAY ICON TOGETHER WITH FREE TEXT				FARP: FUEL AND AMMO	<u>FC</u>		FRIENDLY FIELD ARTILLERY	<u>FL</u>		ENEMY FIELD ARTILLERY	<u>EF</u>
				FORWARD AS-SEMBLY AREA	FA		FRIENDLY FIXED WING	<u>WF</u>		ENEMY FIXED WING	<u>WE</u>
				GRND LIGHTS/ SMALL TOWN	<u>GL</u>		FRIENDLY ATTACK HELI	<u>AH</u>		ENEMY ATTACK HELI	<u>EK</u>
CM				HOLDING AREA	HA		FRIENDLY GEN. ARMY HELI	<u>FG</u>		ENEMY GEN. ARMY HELI	<u>HG</u>
CONTROL MEASURES		51-99 POS 93-99		AIR CONTROL POINT	AP		FRIENDLY HOSPITAL	<u>HO</u>		ENEMY HOSPITAL	<u>EH</u>
	AIRFIELD GENERAL	<u>AG</u>		NDB	<u>BE</u>		FRIENDLY INFANTRY	<u>FI</u>		ENEMY INFANTRY	<u>EI</u>
	AIRFIELD INSTRUMENT	<u>AI</u>		NUC/BIO/CHEM CONTAM. AREA	NB		FRIENDLY MECH. INFANTRY	<u>MI</u>		ENEMY MECH. INFANTRY	<u>EM</u>
	LIGHTED AIRPORT	<u>AL</u>		RAILHEAD: POINT	RH		FRIENDLY MEDICAL	<u>MD</u>		ENEMY MEDICAL	<u>EX</u>
	ARTILLERY FIRING POINT 1	<u>F1</u>		REGIMENT OR GROUP	GP		FRIENDLY TAC OPS CENTER	<u>TF</u>		ENEMY TAC OPS CENTER	<u>ET</u>
	ARTILLERY FIRING POINT 2	<u>F2</u>		US ARMY	US		FRIENDLY UNIT	<u>FU</u>		ENEMY UNIT	<u>EU</u>

TG					
TARGETS, THREATS		T01-T50			
NOT TO BE CONFUSED WITH 'LIVE' ASE THREATS: S6 TSD SYMBOL S6 ASE THREAT		BH BLOODHOUND SAM SYS BH	MK MARKSMAN AIR DEF GUN MK	7 SA-7 SAM SYS 7	SB SABRE AIR DEF GUN SB
		BP BLOWPIPE SAM SYS BP	NV NAVAL AIR DEF SYS NV	8 SA-8 SAM SYS 8	GS SELF-PROP AIR DEF GUN GS
		C2 CSA-2/1/X SAM SYS C2	PT PATRIOT SAM SYS PT	9 SA-9 SAM SYS 9	SP SELF-PROP SAM SYS SP
		CH CHAPPARAL SAM SYS CH	70 RBS-70 SAM SYS 70	10 SA-10 SAM SYS 10	SH SHAHINE R440 SAM SYS SH
T TARGET POINT CURSOR DROP DEFAULTS TO THIS SYMBOL TG	CT CROTALE SAM SYS CT	RA RAPIER SAM SYS RA	11 SA-11 SAM SYS 11	SD SPADA SAM SYS SD	
AD FRIENDLY AIR DEF UNIT AD	GP GEPARD AIR DEF GUN GP	RE REDEYE SAM SYS RE	12 SA-12 SAM SYS 12	SS STARSTREAK SAM SYS SS	
GU GENERIC AIR DEF UNIT GU	G1 GROWTH 1 G1	RO ROLAND SAM SYS RO	13 SA-13 SAM SYS 13	ST STINGER SAM SYS ST	
U UNKNOWN AIR DEF UNIT U	G2 GROWTH 2 G2	1 SA-1 SAM SYS 1	14 SA-14 SAM SYS 14	TR TARGET ACQ RADAR TR	
S6 2S6 / SA-19 AIR DEF UNIT S6	G3 GROWTH 3 G3	2 SA-2 SAM SYS 2	15 SA-15 SAM SYS 15	TC TIGERCAT SAM SYS TC	
AX AMX-13 AIR DEF GUN AX	G4 GROWTH 4 G4	3 SA-3 SAM SYS 3	16 SA-16 SAM SYS 16	GT TOWED AIR DEF GUN GT	
AA AIR DEF GUN AA	HK HAWK SAM SYS HK	4 SA-4 SAM SYS 4	17 SA-17 SAM SYS 17	SA TOWED SAM SYS SA	
AS ASPIDE SAM SYS AS	JA JAVELIN SAM SYS JA	5 SA-5 SAM SYS 5	SM SAMP SAM SYS SM	VU VULCAN AIR DEF GUN VU	
SR BATTLEFIELD SURV RADAR SR	83 M1983 AIR DEF GUN 83	6 SA-6 SAM SYS 6	SC SATCP SAM SYS SC	ZU ZSU-23-4 AIR DEF GUN ZU	

FCR TARGET SYMBOLS							Priority of fire indicators:	
TRACK	WHEEL	AIR DEF / MERGED		UNKN	HELO	FLYER		
LOAL STATIONARY 1500-8000 m								#1 NTS Next-To-Shoot (ready and not)
LOBL STATIONARY 500-1500 m								
LOBL MOVING 500-8000 m								
Time after the last scan when detected targets will turn stale:								stationary: 30 sec
								moving: 5 sec



LEFT CONSOLE

1. STORES JETTISON PANEL
2. INTERIOR LIGHTING PANEL (PRIMARY, SIGNAL LIGHTS TEST)
3. TAIL WHEEL LOCK
4. NVS MODE SWITCH
5. POWER LEVERS
6. EMERGENCY PANEL
7. KU: KEYBOARD UNIT

FRONT PANEL AND RIGHT CONSOLE

8. LEFT MPD (MULTI-PURPOSE DISPLAY)
9. FIRE DETECTION / EXTINGUISHING PANEL (FIRE TESTS)
10. ARMAMENT PANEL (A/S, GND ORIDE)
11. MASTER WARNING / MASTER CAUTION
12. BRU: BORESIGHT RETICLE UNIT
13. EUFD: ENHANCED UP-FRONT DISPLAY
14. RIGHT MPD
15. COMMUNICATIONS PANEL
16. SQL: SQUELCH MOMENTARY TOGGLE SWITCHES
17. RHG: RIGHT HANDGRIP
18. TEDAC: TADS ELECTRONIC DISPLAY AND CONTROL
19. LHG: LEFT HANDGRIP

RADIOS

ADF AN/ARN-149

100 - 2199 kHz
0.1 - 2.199 MHz

CAN DISPLAY BEARING TO ANY DCS 'RADIO TRANSMISSION' TRIGGER THAT IS BROADCASTING WITHIN THE ABOVE RANGE

IDM MD-1295/A

VHF-AM ARC-186

116 - 151.975 MHz
TWO-WAY, NON-SECURE
108 - 115.975 MHz
RECEIVE-ONLY RANGE

UHF-AM ARC-164

225 - 399.975 MHz
DEDICATED GUARD RECEIVER (243 MHz),
HAVEQUICK 2,
KY-58 CIPHER

VHF-FM 2x ARC-201D

FM-1 30 - 87.975 MHz
IFM AMPLIFIER,
SINGGARS
FM-2 30 - 87.975 MHz
SINGGARS

HF ARC-220

2 - 29.9999 MHz
EMBEDDED MODEM,
KY-100 CIPHER

XPNDR APX-118

MODES 1, 2, 3/A,
4, C, and S

INTERCOM

1) RADIO SELECTED FOR IDM DATA:

- BY THIS CREW MEMBER
- BY THE OPPOSITE CREW MEMBER
- BY BOTH CREW MEMBERS

2) RADIO SELECTED FOR TALKING:

- BY THIS CREW MEMBER
- BY THE OPPOSITE CREW MEMBER
- BY NEITHER (LISTENING-ONLY)

3) RADIO & ITS *SQUELCH STATUS

4) ACTIVE FREQUENCY AND MODE:

- F FREQUENCY HOPPING
- U UPPER SIDEBAND
- L LOWER SIDEBAND
- C CONTINUOUS WAVE
- A AMPLITUDE MODULATION EQUIVALENT

5) CALLSIGN AND CIPHER STATUS

6) GUARD AND IFM POWER STATUSES

EUFD

1	2	3	4	5	6	7	8
UHF*	121.500	SABER	L2	121.500	----		
UHF*	305.000	A2L86 C2 G	T8	305.000	----		
FM1*	F123	----	NORM	30.000	----		
FM2*	30.000	----	L6*	30.000	----		
HF*	2.000 A	----	LOW	2.000 A	----		
FUEL 1780			XPNDR S 1200 A	NORM	04:21:54 Z		

7) IDM NET STATUS:

- I INTERNET
- F FIRE SUPPORT
- L LONGBOW
- T TACFIRE
- * CONNECTED TO TAC INTERNET

8) STANDBY FREQ

- 9) XPNDR STATUS
- 10) MODE 4 CODE
- 11) MODE 3/A CODE
- 12) MODE S ENABLED

LEFT CONSOLE

1. FAT: FREE AIR TEMPERATURE GAGE
2. EXTERIOR / INTERIOR LIGHTING PANEL (PRIMARY, ANTI-COL, SIGNAL LIGHTS TEST)
3. STORES JETTISON PANEL
4. ENGINE START MOMENTARY TOGGLE SWITCHES
5. APU START/STOP
6. POWER LEVERS
7. MSTR IGN SWITCH
8. RTR BRK SWITCH
9. EMERGENCY PANEL
10. NVS MODE SWITCH
11. TAIL WHEEL LOCK
12. KU: KEYBOARD UNIT

FRONT PANEL AND RIGHT CONSOLE

13. VIDEO PANEL (ACM)
14. ARMAMENT PANEL (A/S, GND ORIDE)
15. LEFT MPD (MULTI-PURPOSE DISPLAY)
16. FIRE DETECTION / EXTINGUISHING PANEL (FIRE TESTS)
17. STANDBY COMPASS
18. MASTER WARNING / MASTER CAUTION
19. BRU: BORESIGHT RETICLE UNIT
20. EUFD: ENHANCED UP-FRONT DISPLAY
21. CMWS: AN/AAR-57 COMMON MISSILE WARNING SYSTEM
22. RIGHT MPD
23. STANDBY ATTITUDE INDICATOR
24. STANDBY ALTIMETER
25. COMMUNICATIONS PANEL
26. SQL: SQUELCH MOMENTARY TOGGLE SWITCHES
27. DTU: DATA TRANSFER UNIT
28. STANDBY AIRSPEED INDICATOR
29. GEN RST SWITCH
30. PARK BRAKE HANDLE

QUICKSTART

- 30 SET (HANDLE OUT) 7 TO BATT
- 5 APU ON
- 20 WAIT 20 SEC FOR 'APU ON'
- CLOSE THE CANOPY (LCTRL + C)
- 25 RLWR TO MAX
- 26 FLIP FORWARD ALL SQL SWITCHES
- 21 PWR ON, ARM, CMWS
- 15 [M] > [B2 ENG] AND MONITOR Ng

4 NO.1 TO START AND WAIT 3-5 SEC

6 NO.1 TO IDLE (RALT + HOME)

WAIT 60 SEC UNTIL Ng1 IS AT 66-67%

4 NO.2 TO START AND WAIT 3-5 SEC

6 NO.2 TO IDLE (RSHIFT + HOME)

WAIT 50 SEC UNTIL Ng2 IS AT 66-67%

6 BOTH SMOOTHLY TO FLY (NUMPAD +)

WAIT UNTIL Nr AND Np ARE AT 101%

5 APU OFF

PERFORM IHADSS BORESIGHTING:

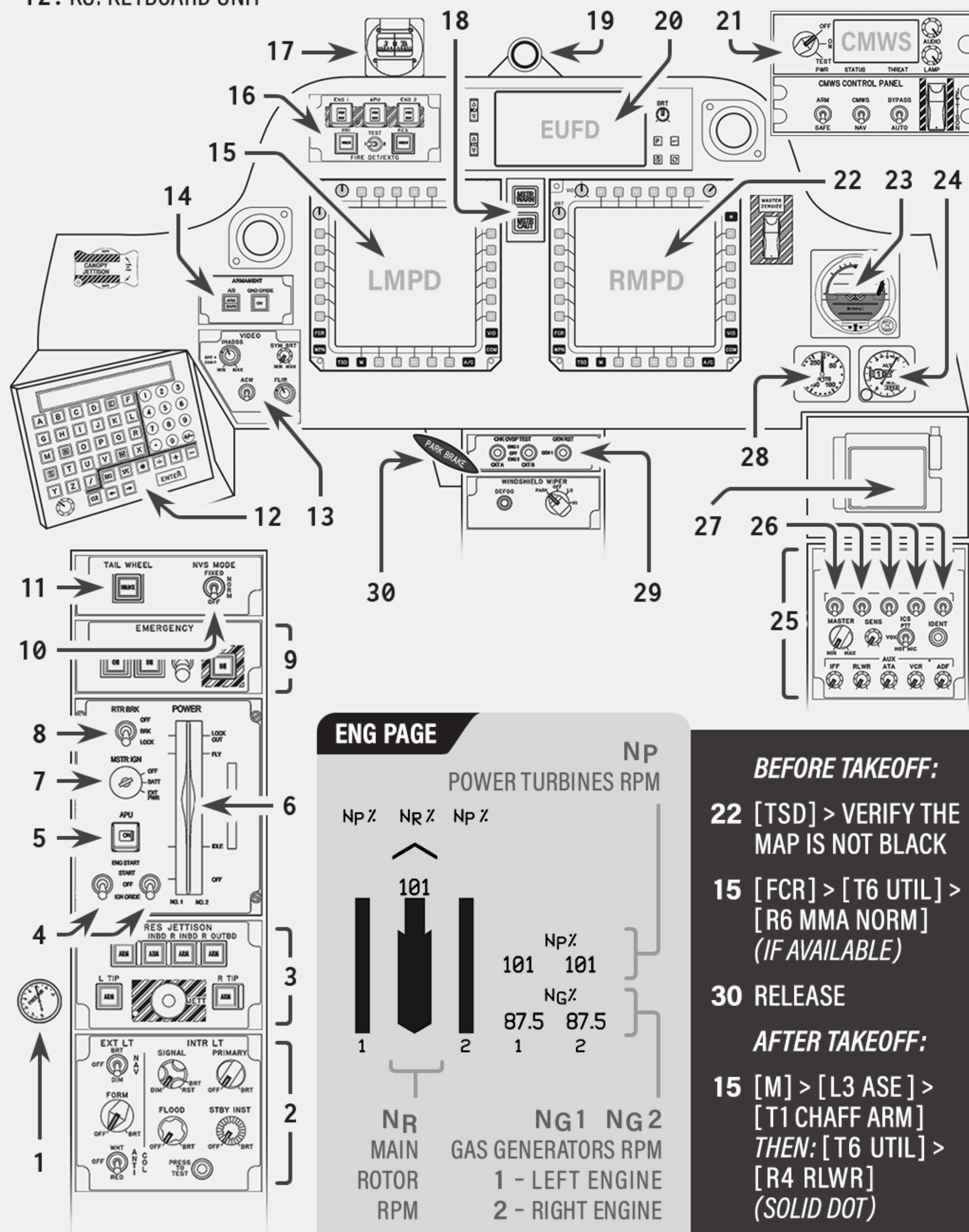
15 [WPN] > [L5 BORESIGHT] > [L4 IHADSS]

2 PRIMARY KNOB TO BRT

19 MOVE THE CAMERA TO ALIGN HMD RETICLE WITH BRU:



15 [L6 B/S NOW] TO CONFIRM



BEFORE TAKEOFF:

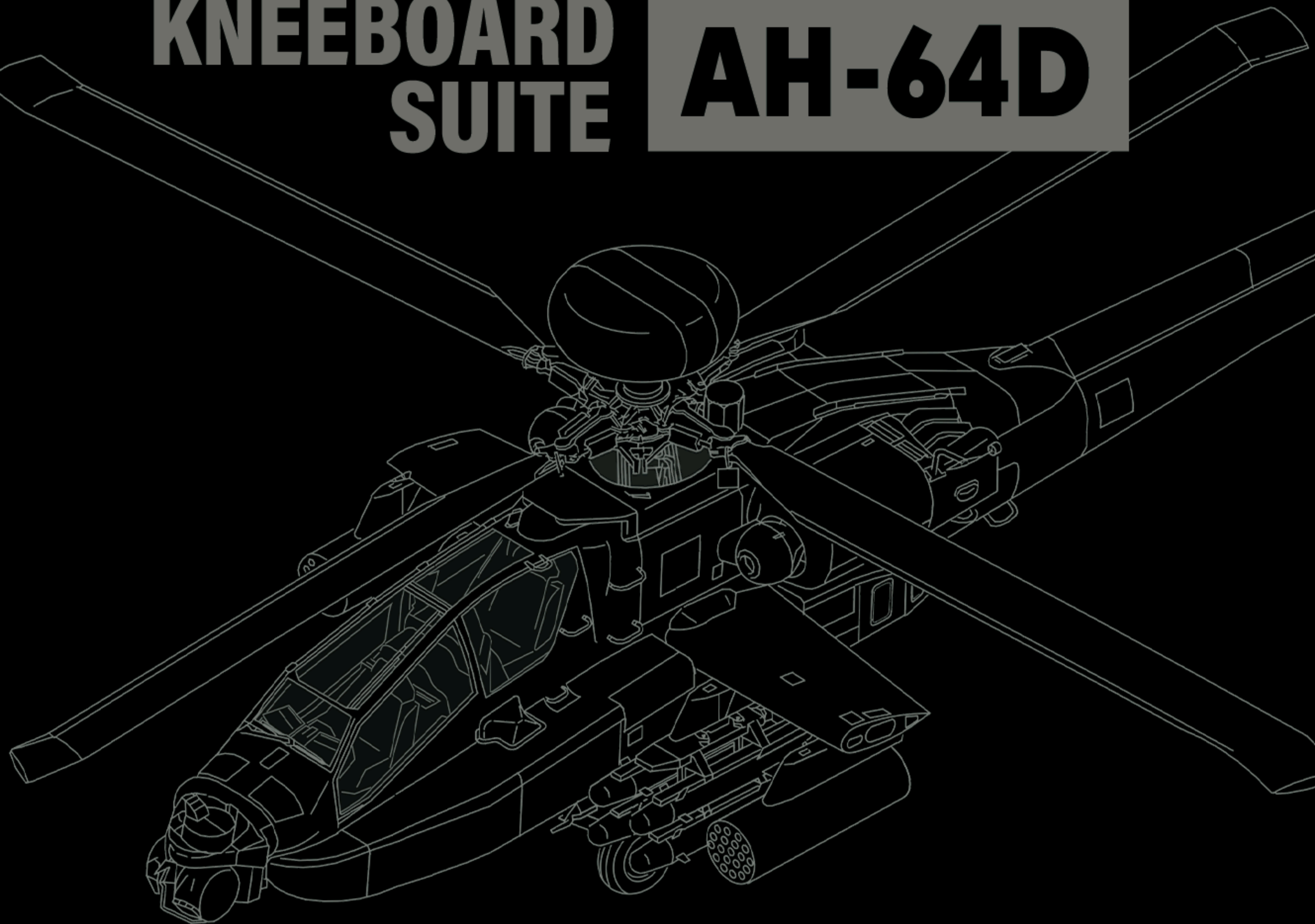
- 22 [TSD] > VERIFY THE MAP IS NOT BLACK
- 15 [FCR] > [T6 UTIL] > [R6 MMA NORM] (IF AVAILABLE)
- 30 RELEASE

AFTER TAKEOFF:

- 15 [M] > [L3 ASE] > [T1 CHAFF ARM] THEN: [T6 UTIL] > [R4 RLWR] (SOLID DOT)

KNEEBOARD SUITE

AH-64D



CHECKLISTS, Pt. 1. DMS SWEEP.

2

CHECKLISTS, Pt. 2. WAILRM. COORDS.

3

ASE THREATS

4

GEORGE AS CPG

5

GEORGE AS PILOT

6

ORDNANCE. REARMING.

7

REFUELING. FUEL TRANSFER.

8

TSD SYMBOLS, Pt. 1

9

TSD SYMBOLS, Pt. 2. FCR TARGETS.

10

CPG STATION. RADIOS.

11

PILOT STATION. QUICKSTART.

12

PLT	INTERIOR CHECK
1	EXT / INTR lights, COMM panel, KU, MPD, and EUFD knobs As desired
2	PARK BRAKE <i>handle out</i> Set
3	POWER levers, RTR BRK switch, NVS MODE switch OFF

CPG	INTERIOR CHECK
1	INTR lights, COMM panel, TEDAC, KU, MPD, and EUFD knobs As desired
2	POWER levers, NVS MODE switch OFF

ONCE BATTERY POWER IS AVAILABLE:
SIGNAL LIGHTS TEST. FIRE TESTS 1 & 2.

PLT	STARTING APU
BATTERY TIME UNDER LOAD IS LIMITED BY FAT TO: >43°C = 1 MIN 32-43°C = 3 MIN <32°C = 20 MIN	
1	MSTR IGN switch <i>rmb</i> BATT
2	TAIL WHEEL Verify locked
SIGNAL LIGHTS TEST. FIRE TESTS 1 & 2.	
3	APU push button Press <i>'APU ON' EUFD advisory will appear within 20 sec</i>

PLT	AFTER STARTING APU
1	Canopy door Closed
2	RLWR volume As desired
3	SQL momentary switches <i>rmb</i> Flip forward
4	
5	DMS sweep Perform
6	CMWS Control Panel As desired
7	NVS MODE switch As desired
8	SAI <i>scroll down</i> Uncaged

CPG	AFTER STARTING APU
1	Canopy door Closed
2	DMS sweep Perform

IHADSS BORESIGHTING

- 1 [SIGHT SELECT SWITCH] — to HMD
- 2 WPN > BORESIGHT page > IHADSS
- 3 INTR LT > Adjust PRIMARY knob
so that the symbol on BRU is visible
- 4 Align HMD reticle with BRU
- 5 Press B/S NOW to confirm position



THE SWEEP ITEMS ARE SPLIT AS BRIEFED. TYPICALLY, THE PILOT
TAKES COMMS AND NAVIGATION, AND THE CPG TAKES WEAPONS
AND SENSORS. THE SWEEP MAY BE FINISHED AFTER TAKEOFF.

DMS SWEEP

ASE > UTIL

- ▶ RLWR — ON
- ▶ Choose CHAFF MODE
and adjust PROGRAM
- ▶ Adjust flare program
(via the kneeboard)

TSD > SHOW

- Configure SHOW,
THRT SHOW, and
COORD SHOW for
NAV and ATK phases

TSD > UTIL

- TIME — ZULU or LOCAL
- Verify SYSTEM TIME
- DOPPLER — verify ON

TSD > POINT

- Check/add points
and targets

TSD > RTE

- Check/adjust route,
set DIR as needed

TSD > BAM

- Set up Priority Fire
and No-Fire zones

TSD > INST > UTIL

- ▶ ADF — ON

WPN > CODE > FREQ

- Modify laser codes
as needed. For example:
A — freq for self-lasing,
B — wingman's freq,
C — JTAC's freq.

WPN > CODE

- Assign laser codes
as needed. For example:
LRFD — A, LST — B or C.

WPN > CHAN

- Assign laser codes to
the missile channels
as needed, or leave as is:
1 — A, 2 — B, 3 — C.

WPN > UTIL < PILOT

- ▶ IHADSS, PNVS — ON

WPN > UTIL < CPG

- ▶ IHADSS, TADS, FLIR,
and LASER — ON

WPN > MANRNG

- ▶ '800', 'A', or as desired

WPN > GUN

- ▶ Select BURST LIMIT

WPN > MSL

- ▶ If self-lasing is desired:
PRI code = LRFD code

WPN > RKT

- ▶ Select INVENTORY
type and QTY

WPN > BORESIGHT

- ▶ IHADSS < PILOT < CPG

FCR > UTIL

- ▶ MMA (if avail.) — NORM

FLT > SET

- ▶ RDR ALT — verify ON
- ▶ Set HI/LO alt alerts
- ▶ Set ALT to field
elevation
or set PRESS to QNH
- Set standby altimeter
- UNITS — as desired
- Horizon line — as desired

FUEL

- AUX GALLONS EXT:
enter the total amount of
fuel in all external tanks
(230, 460, 690, 920)

UTIL

- After closing both doors:
ECS — verify ON, temp
set as desired (50—90°F)

COM

- ▶ Radios — check/tune

- UHF GUARD REC — ON

- XPNDR — as desired *MP*

- Verify ORIG ID
and CALL SIGN

- Set up datalink preset:
RETRIES 0, TUNE SC,
select on the EUFD, etc.

- Set up team members

- Test datalink: TSD > RPT >
PP > MSG RQST > SEND

PLT	STARTING ENGINES
1	Control sweep, trim check Perform
2	ANTI-COL lights WHT (<i>day</i>), RED (<i>night</i>) or OFF (<i>when required</i>)
3	First Engine START switch <i>rmb</i> START
4	At first indication of NG increase: POWER lever IDLE
5	Once first engine is stabilized at 66–67% NG: Second Engine START switch <i>rmb</i> START
6	At first indication of NG increase: POWER lever IDLE
7	Once both engines are stablilized: POWER levers Advance smootly to FLY
8	Once Np and NR are stabilized at 101%: APU Off

BEFORE TAXI CHECK	
1	TSD page ENSURE THE MAP IS DISPLAYED (EGI "ALIGNMENT" TAKES 5 MIN WITH OR WITHOUT GPS)
2	Chocks Removed
3	 As desired
4	PARK BRAKE Release
5	TAIL WHEEL Unlock
6	Wheel brakes (<i>both seats</i>) Check during taxi

BEFORE TAKEOFF CHECK	
1	Weapons Not actioned, SAFED, ORIDE off
2	TAIL WHEEL button Locked, no light
3	PARK BRAKE As desired
4	POWER levers To FLY
5	XPNDR As desired
6	BINGO or ADF timer As desired
7	Power check Perform (<i>validate PERF page</i>)

LEVEL OFF CHECK	
1	WAILRM–NCA Perform
2	FUEL page > AUX transfer As desired
3	DMS sweep (<i>if not already done</i>) Finish

BEFORE LANDING CHECK	
1	Weapons Not actioned, SAFED, ORIDE off
2	ASE & CMWS Control Panel As required
3	TAIL WHEEL Locked
4	PARK BRAKE Released

PLT	ENGINE SHUTDOWN (APU)
IF ENGINES ARE NOT IDLED FOR 2 MINUTES PRIOR TO SHUTDOWN, A RESTART SHOULD BE AVOIDED BETWEEN 5 MINUTES AND 4 HOURS.	
1	APU button Press ON
2	TAIL WHEEL / PARK BRAKE Lock / Set
3	After 'APU ON' EUFD advisory is displayed: POWER levers IDLE
4	SAI <i>lmb</i> + scroll up Cage
5	NVS MODE switch OFF
6	ACM switch VIDEO panel OFF
7	WPN > UTIL page > PNVS OFF
8	After idling for 2 minutes: POWER levers OFF
9	Once below 50% NR: RTR BRK switch BRK
10	Stabilator (ENG > SYS > STAB) Set to ZERO
11	When rotor stops: RTR BRK switch OFF
12	EXT / INTR lights, Searchlight OFF
13	APU button OFF
14	MSTR IGN switch OFF
15	Before leaving the helicopter Set chocks

CPG	ENGINE SHUTDOWN (APU)
1	NVS MODE switch OFF
2	TEDAC display knob OFF
3	DMS > SHUTDOWN page > MASTER OFF
4	INTR lights OFF

(W) WEAPONS

(A) ASE

- CMWS
- CHAFF
- RLWR

(I) IFF

(L) LIGHTS

(R) RECORDER

(M) MPDs, ACQ

(N) NAV

(C) COMMS

(A) AIR SURV

- C-SCOPE

UTM > 4+4 MGRS

TSD DISPLAY FORMAT:
47 00A BC 1234 5678
FOUR-DIGIT EASTING & NORTHING

KU ENTRY FORMAT:
00ABC12345678
TO CHANGE DATUM (47): TSD > UTIL

LAT/LONG > DD°MM.mm

DCS F10 MAP FORMAT:
N12°34.567 E12°34.567
LLDM (LAT LONG DECIMAL MINUTES)

TSD DISPLAY FORMAT:
N12 34.56 E012 34.56
DEGREES OF LONGITUDE IS ALWAYS THREE-DIGIT (ADD LEADING ZERO).
DECIMALS ARE ALWAYS TWO-DIGIT.

KU ENTRY FORMAT:
N123456E0123456

	SURFACE	NAVAL	MISSILES		nm / km
2	SA-2 Guideline S-75	Amazonka RF	/ Fan Song TR	C	19/35
3	SA-3 Goa S-125		Low Blow TR	C	08/15
5	SA-5 Gammon S-200		Square Pair TR	C	135/250
6	SA-6 Gainful Kub		Straight Flush STR	C	16/30
7	HQ-7B		SR / TELAR	E	07/13
8	SA-8 Gecko Osa		TELAR	C	06/11
10	SA-10 Grumble S-300PS:				
		Big Bird SR	/ Clam Shell SR / Flap Lid A & B TR	C	25/46
11	SA-11 Gadfly Buk	Snow Drift SR	/ Fire Dome TELAR	C	16/30
13	SA-13 Gopher Strela-10M		TELAR	F	02/04
15	SA-15 Gauntlet Tor		TELAR	C	07/13
	SA-15 Gauntlet M2 Tor-M2		TELAR	C	08/15
22	SA-22 Greyhound Pantsir-S1		SPAAW	C	11/20
AA	Fire Can SON-9	AAA Fire Director Radar			
GS	Gepard		SPAAA	E	03/06
GS	LPWS Centurion Phalanx		C-RAM	E	02/04
GS	Vulcan		SPAAA	E	02/04
HK	Hawk		CWAR / SR / TR	C	11/20
IT	IRIS-T SLM		STR	F	22/41
MM		MISSILE			
NV		NAVAL UNIT			
PT	Patriot		STR	C	23/42
RA	Rapier with Blindfire Radar		TR	E	05/10
	Rapier with or without optical Tracker		LN	E	05/10
RO	Roland		EWR / ADS	C	06/11
S6	SA-19 Grison Tunguska		SPAAW	E	05/10
SA	NASAMS with AIM-120B		SR	C	07/13
	NASAMS with AIM-120C		SR	C	08/15
SR	Nebo 1L13 / Nebo 55G6 / AN/FPS-117		EWR		
TR	Dog Ear Sborka	[13-15-19]	MCC-SR		
	Flat Face B	[2-3-5]	SR		
	Tin Shield	[5-10]	SR		
U	HQ-7 'Player' AI- or user-controlled		TELAR	E	07/13
	Freya / Würzburg	WWII Assets Pack	EWR		
	Silkworm	anti-ship	SR		
	Tall King	[5]	SR		
ZU	ZSU-23-4 Shilka		SPAAA	E	02/04

	AIRBORNE	
14	F-14A / GR/B	123
15	F-15C	123
	F-15E	23
	F-15E S4+	123
16	F-16A	123
	F-16A MLU	23
	F-16C / CM	23
18	F/A-18A	12
	F/A-18C	123
19	MiG-19P	2
21	MiG-21Bis	12
23	MiG-23MLD	12
25	MiG-25PD	12
29	MiG-29A/G	12
	MiG-29S	123
	Su-27/33	12
	J-11A	123
30	Su-30	123
31	MiG-31	12
34	Su-34	123
37	AJS-37 Viggen	2
50	A-50 / KJ-2000	AWACS
E2	E-2D	AWACS
E3	E-3A	AWACS
F1	Mirage F1	12
F2	Tornado GR4	2
F4	F-4E-45MC	12
F5	F-5E / E-3 / FC	2
JF	JF-17	23
M2	M-2000C	12
	Mirage 2000-5	123
U	A-6E	2
	AH-64D (old)	3
	C-101CC	2
	H-6J	BOMBER
	Tornado IDS	2
	Tu-95MS	BOMBER
	SH-60B	HELO

1 — SARH, 2 — IR, 3 — ARH — SUPPORTED A-A MISSILES. CM TO USE: C — CHAFF, F — FLARES, E — EVADE.
19/35 — SAM'S MAX FIRING RANGE vs. AH-64D AT 3000 ft, nm / km.
[THE SAM] TO WHICH THIS RADAR IS USUALLY LINKED.
IN GREEN — SUPPORTING UNIT: DESTROYING IT MIGHT NOT DISABLE THE SITE.

XX

SURFACE THREAT

XX

AIRBORNE THREAT

XX

ACQUISITION: BOXED ICON

XX

TRACKING: BOXED + DASHED LINE

XX

LAUNCH: FLASHING BOX

*

LWR THREAT

*

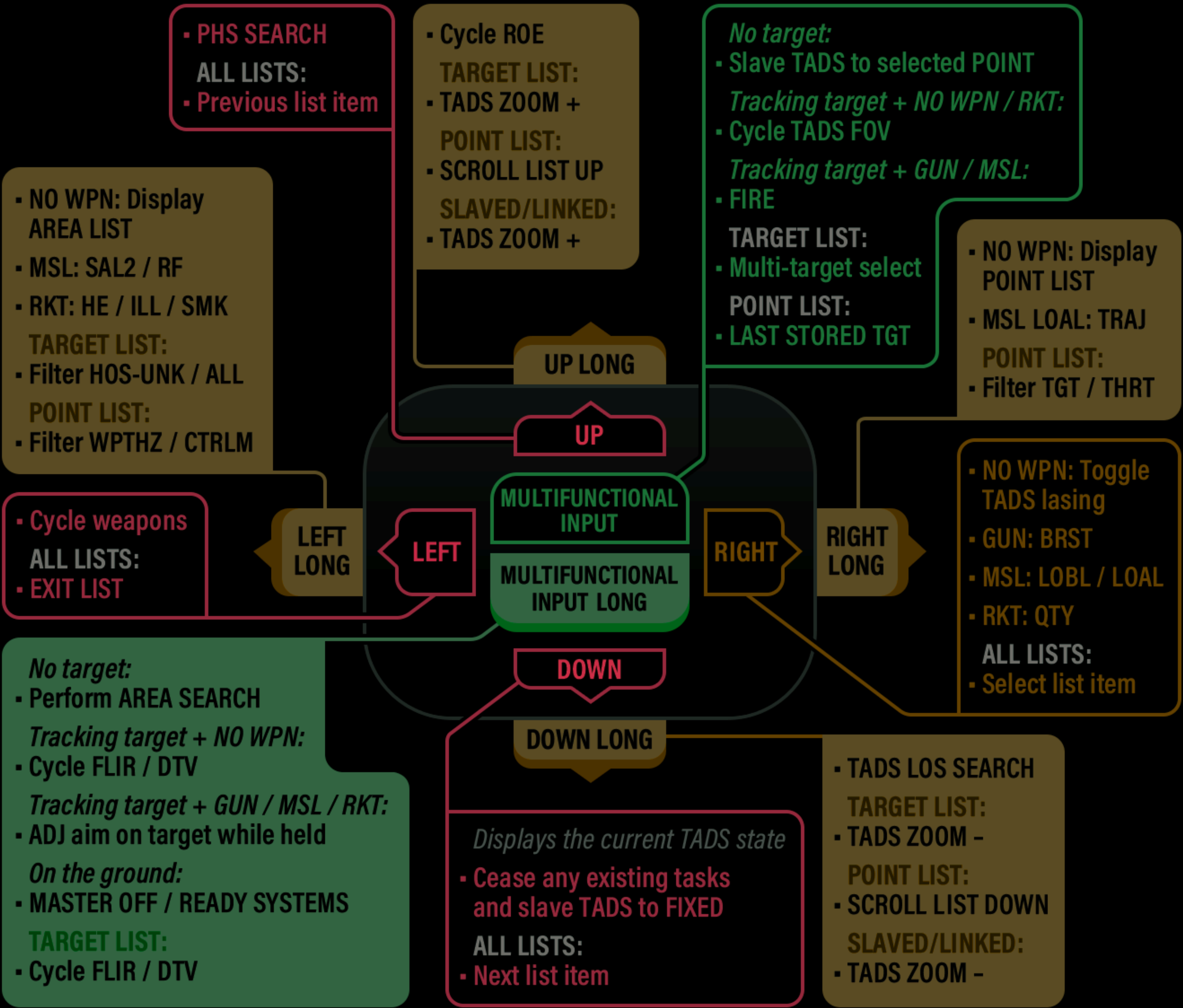
RANGING: BOXED ICON

*

DESIGNATING: BOXED + DASHED LINE

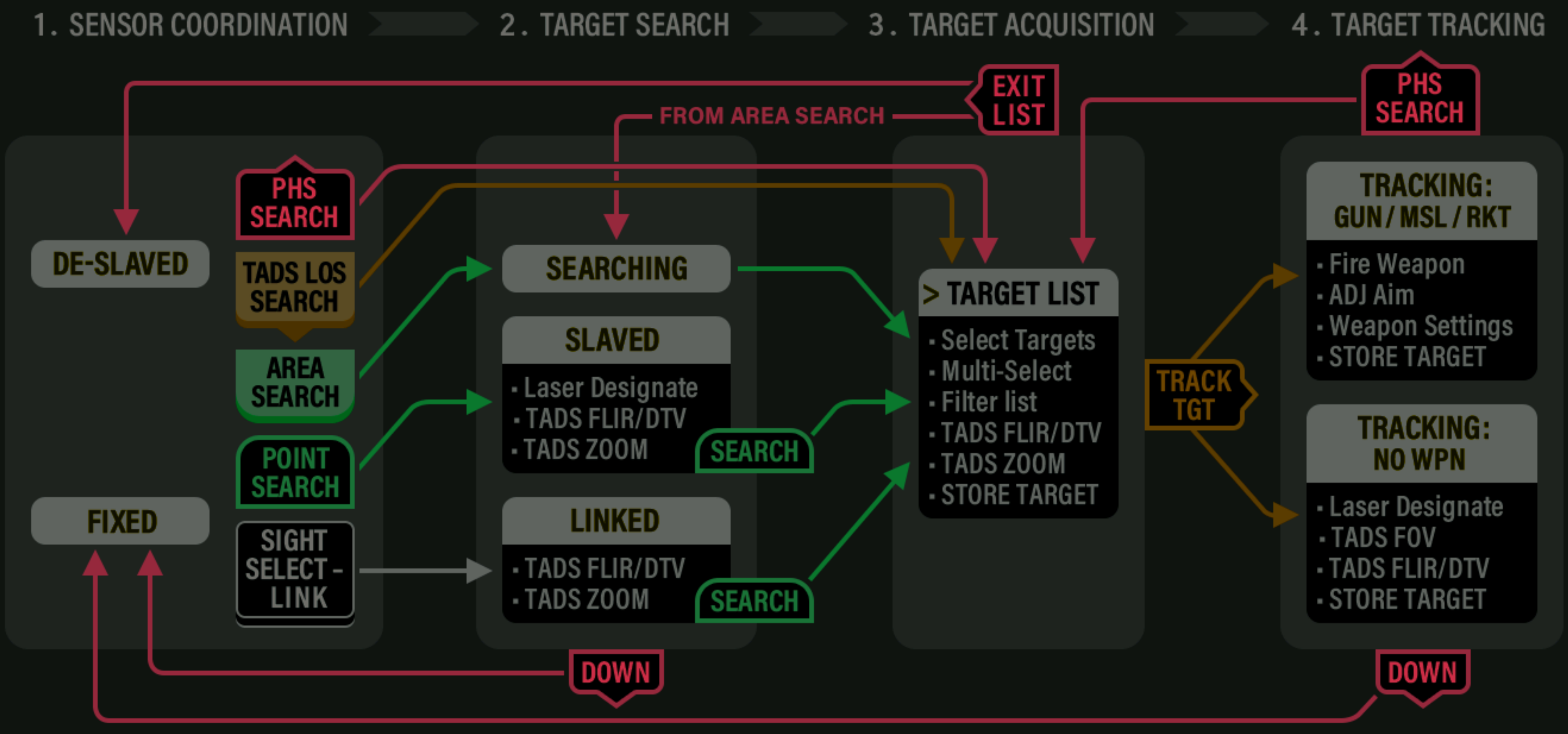
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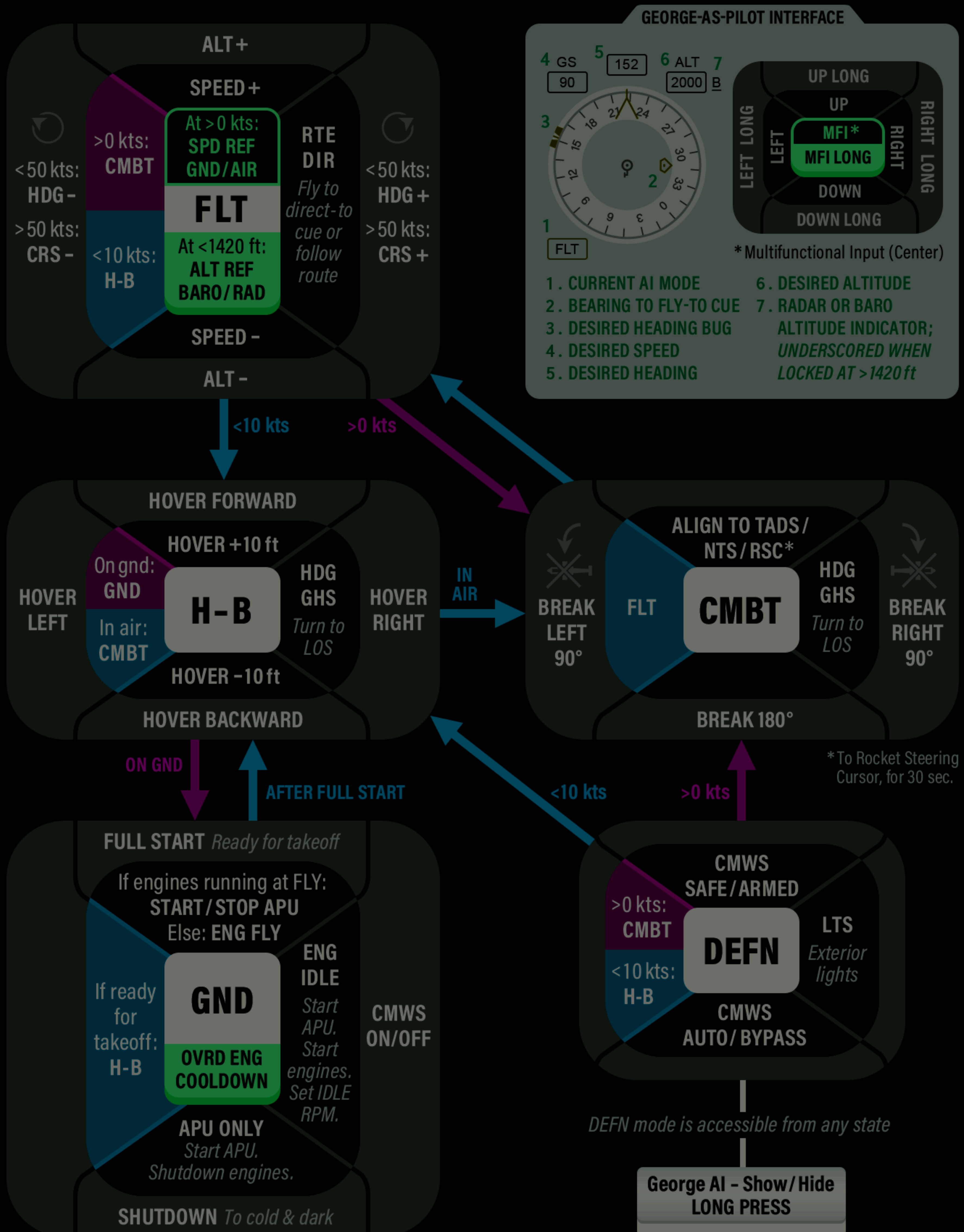
BEAMING: FLASHING BOX



NORMAL ACTIONS (IN BROWN) AND COORDINATED ACTIONS (IN GREEN) ARE BROADLY CONTEXTUAL DEPENDING ON THE CURRENT TASK. PRIORITY ACTIONS (IN RED) ARE CONSTANT.

GEORGE-AS-CPG WORKFLOW





RAPID REARMING

- 1 Prior to entering the FARP:
Weapons SAFED, ORIDE OFF
- 2 WPN > UTIL page:
MSL, RKT, GUN OFF
- 3 TAIL WHEEL Locked
- 4 PARK BRAKE Set
- 5 IDM and HF:
Auto transmit Disable
- Rearming completed:
- 6 WPN > UTIL page:
MSL, RKT, GUN ON
- 7 FUEL page: 230/460/690/920
AUX GALLONS EXT Update

NM = KM ÷ 1.85

M = FT ÷ 3.28

KM = NM × 1.85

FT = M × 3.28

1 KM = 0.54 NM

LBS = GAL × 6.66

GROUND SPEED REQUIRED (KTS) =
(DISTANCE ÷ MINUTES) × 60

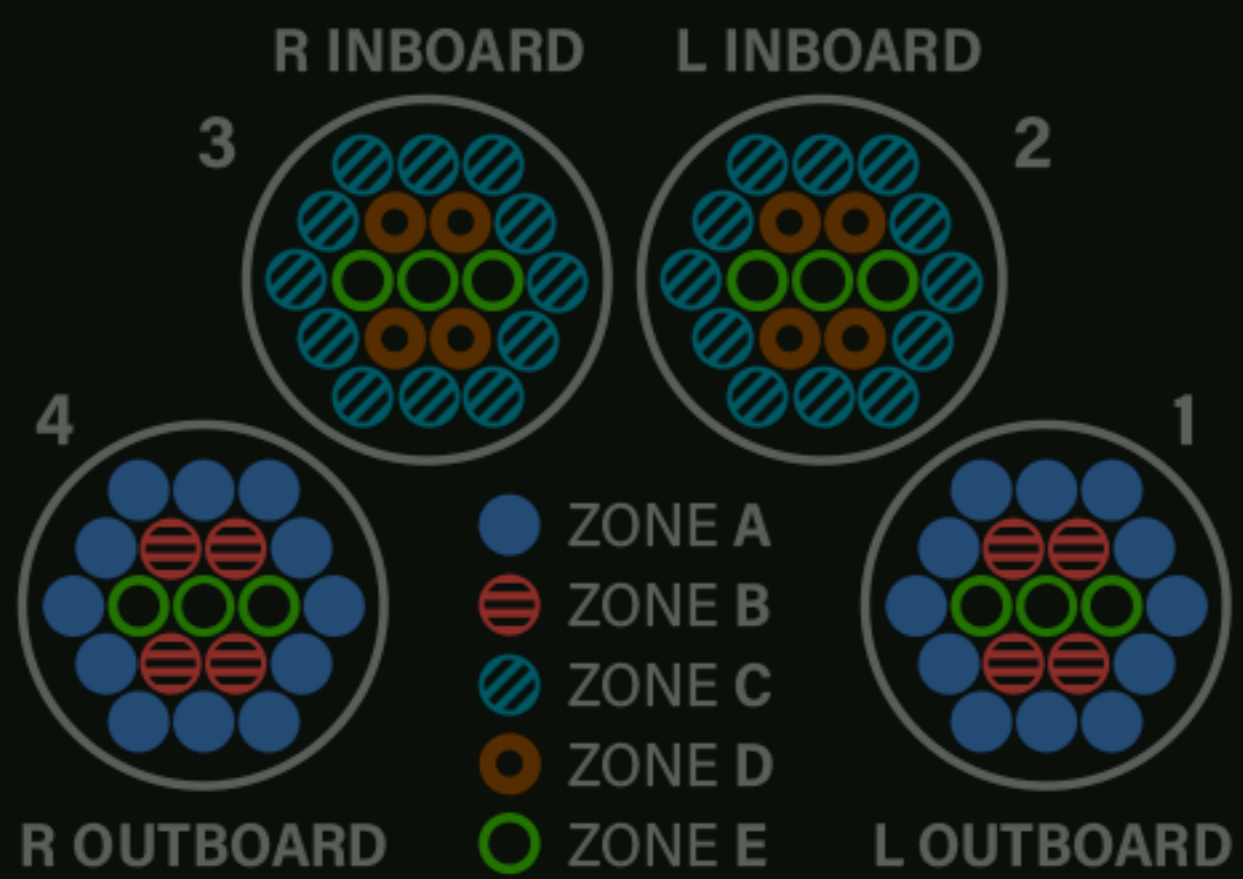
TIME OF FLIGHT (MINS) =
(DISTANCE ÷ GROUND SPEED) × 60

BINGO FUEL (LBS) =
(TIME OF FLIGHT ÷ 60) × FUEL LB/HR

OBJECTIVE TIME (MINS) =
([TOTAL FUEL - BINGO FUEL]
÷ FUEL LB/HR) × 60

SFR (SPECIFIC FUEL RANGE) FACTOR =
GROUND SPEED ÷ FUEL LB/HR

FLIGHT RANGE (NM) =
SFR × TOTAL FUEL



EMPTY WEIGHT (NO MMA)	13074
MAX WEIGHT	23000
MAX FUEL CARRIED:	
FWD INTERNAL TANK	1040
AFT INTERNAL TANK	1460
IAFS (ROBBIE TANK)	650
EXTERNAL TANKS	4x 1525

U.S. Army AH-64D Block 2 (2005-2010)
Lot 9.1 avionics, T-700-GE-701C engines

524 Mast Mounted Assembly
FCR/RFI

0 30 chaff + 60 flares
non-editable

980 98 GAL IAFS (ROBBIE) C AUX
Limits gun ammo to 300 rounds

M230E1 30mm gun

• M789 HEDP high explosive

• M788 TP target practice

227 Up to 300 rounds (with IAFS)

908 Up to 1200 rounds (w/o IAFS)

A/G MISSILES

144 M299 Hellfire Launcher

100 AGM-114K Hellfire SAL

108 AGM-114L Longbow Hellfire RF

M261 ROCKET PODS

SINGLE TYPE HYDRA 70/Mk 66

523 19x M151 HE 6PD
10 lb warhead, soft targets

523 19x M151 HE, M433 RC 6RC
custom delay, soft targets

668 19x M229 HE 6PD
17 lb warhead, soft targets

556 19x M257 IL 6IL
illumination flare (3 min),
deployed 3500 m
from the launch point

523 19x M274 TP-SM 6SK
target practice

607 19x M282 MPP 6RC
penetrator, light armor

MIXED LOAD, INBOARD

527 C: 12x M257 IL 6IL
D/E: 7x M151 6PD

522 C: 12x M274 TP-SM 6SK
D/E: 7x M151 6PD

MIXED LOAD, OUTBOARD

527 A/B: 16x M151 6PD
E: 3x M257 IL 6IL

522 A/B: 16x M151 6PD
E: 3x M274 TP-SM 6SK

FUEL TANKS

1690 230 GAL EXT TANK L AUX R AUX
165 lbs empty tank,
1525 lbs of fuel,
no ballistic protection,
no quantity probes

DCS PODS PAIRING LOGIC:

WHEN INSTALLING
OUTBOARD PODS ONLY,
THEY MUST BE THE SAME.

WHEN INSTALLING
INBOARD PODS ONLY,
THEY MUST BE THE SAME.

WHEN INSTALLING OUTBOARD
AND INBOARD PODS TOGETHER,
THEY MUST EITHER BE THE SAME,
OR ALL HAVE THE EXACT SAME
ROCKET (INCLUDING THE FUZE)
IN THEIR ZONE E.

NOTE: WITHOUT THE INBOARD
TANK INSTALLED, THE FUEL IN
ITS ADJACENT OUTBOARD
TANK WILL BE INACCESSIBLE.

RAPID (HOT) REFUELING

ENGINE 2 SHUT DOWN, OR BOTH OPERATING

- 1 Prior to entering the FARP:
Weapons SAFED, GND ORIDE off
- 2 TAIL WHEEL / PARK BRAKE Locked / Set
- 3 POWER levers:
A Both engines running:
1) POWER lever **NO 2** IDLE
2) FUEL page > XFER OFF
B Single engine running:
1) POWER lever **NO 2** IDLE for 2 minutes, then OFF
- 4 IDM and HF auto transmit Disable
- 5 ANTI-COL switch OFF
- Refueling completed:
- 6 IDM and HF auto transmit As required
- 7 ANTI-COL switch As desired
- 8 FUEL page > AUX GALLONS EXT Update

NON-SANDY ENVIRONMENT

- 9 (Bubble Burn) Apply collective until 60% torque is reached or aircraft is light on wheels. Maintain power settings for 60 sec.
- 10 Collective Reduce to minimum torque
- 11 POWER levers:
A Both engines running:
1) POWER lever **NO 2** To FLY
2) FUEL page > XFER AUTO
B Single engine running (cross-bleed start):
1) Collective Increase to attain 95% NG2
2) **Engine 2** Start
3) Collective Reduce to flat pitch after starter dropout
4) POWER lever **NO 2** To FLY

RAPID (WARM) REFUELING

BOTH ENGINES SHUT DOWN AND APU ON

- 1 Prior to entering the FARP:
Weapons SAFED, GND ORIDE off
- 2 TAIL WHEEL / PARK BRAKE Locked / Set
- 3 APU Start (wait for 'APU ON' on EUFD)
- 4 POWER levers IDLE for 2 minutes, then OFF
- 5 IDM and HF auto transmit Disable
- 6 Once below 50% NR:
RTR BRK switch BRK until rotor stops, then OFF
- 7 ANTI-COL switch OFF
- Refueling completed:
- 8 IDM and HF auto transmit As required
- 9 FUEL page > AUX GALLONS EXT Update
- 10 Perform STARTING ENGINES procedure **Pt.2**

SANDY ENVIRONMENT

- 9 POWER levers:
A Both engines running:
1) POWER lever **NO 2** To FLY
2) FUEL page > XFER AUTO
B Single engine running:
1) APU Start
2) **Engine 2** Start
3) POWER lever **NO 2** To FLY
- 10 FUEL page > CROSSFEED Switch:
1) To FWD, wait 60 sec.
2) To NORM, wait 30 sec.
3) To AFT, wait 60 sec
4) To NORM, wait 30 sec.

FUEL TRANSFER

WHEN TO BEGIN FUEL TRANSFER

FROM C AUX:

Typically, after the second 30-minutes BINGO alert (FUEL > CHECK), or after the TOT drops below 2500 lbs.

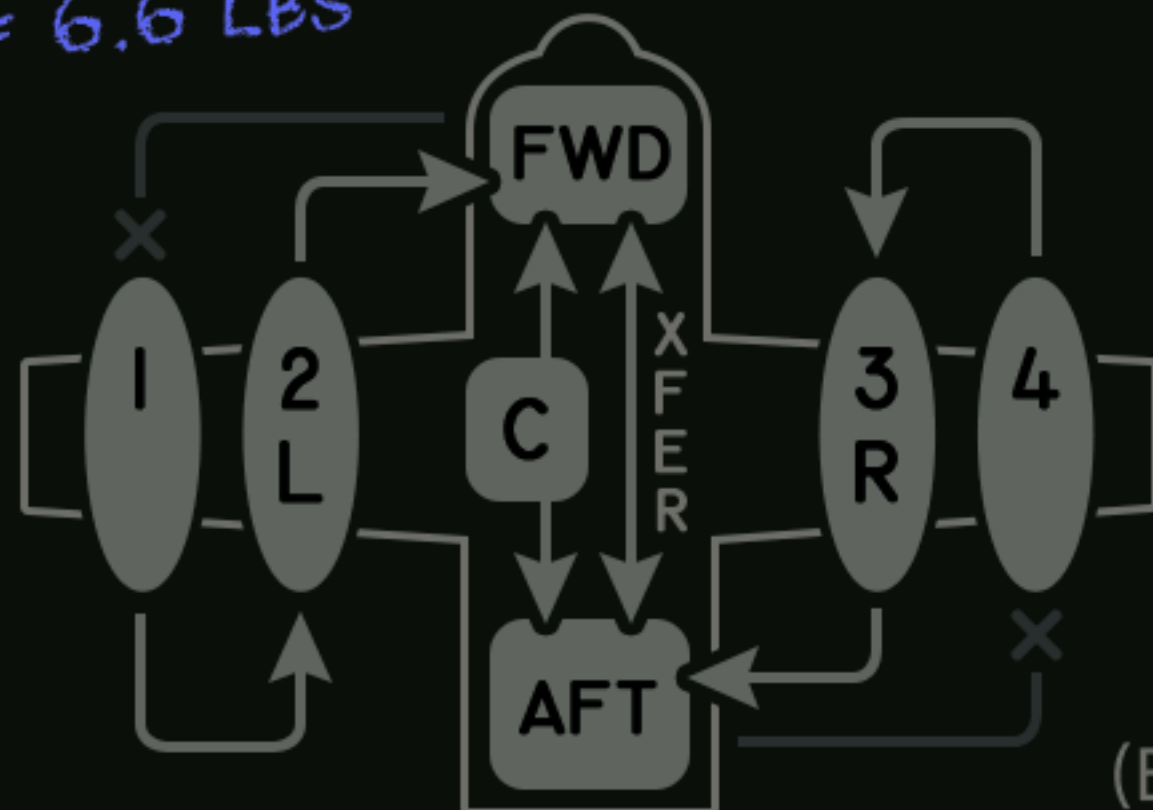
FROM EXT AUX:

Typically, after takeoff.

FUEL TRANSFER PRIORITY:

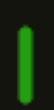
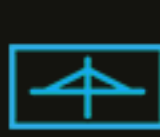
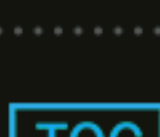
- 1. FWD / AFT (XFER)
 - 2. EXT > FWD / AFT
 - 3. C AUX > FWD / AFT
- Meaning, transfer from the C AUX will not start until the FWD / AFT / EXT transfer is completed or suspended.

1 GAL = 6.6 LBS



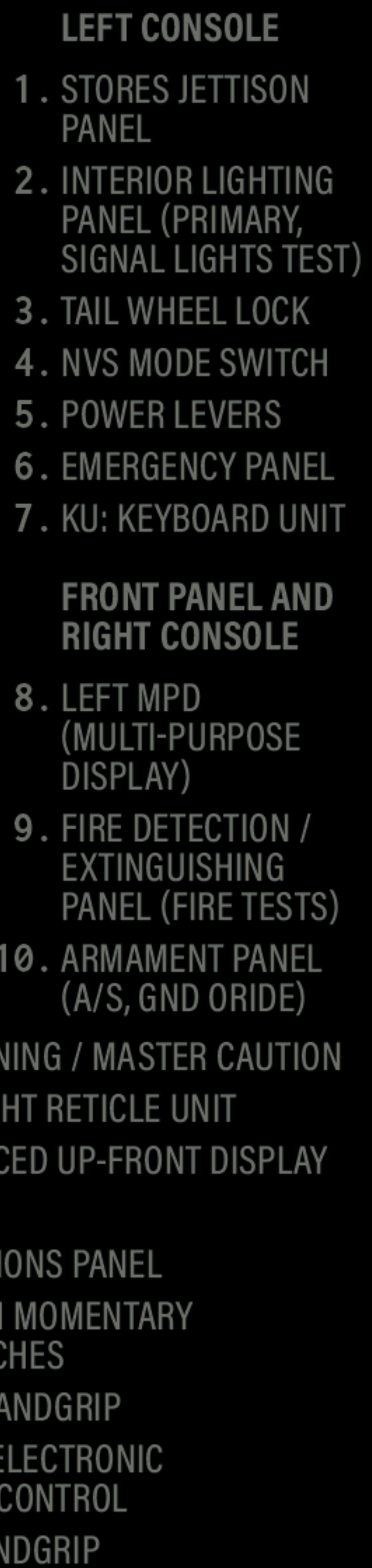
FWD	1040 lbs
	156 gal
AFT	1460 lbs
	220 gal
C AUX	650 lbs
(ROBBIE)	98 gal
EXT AUX	1525 lbs
(EACH TANK)	230 gal

NOTE: THE **OUTBOARD** EXT TANKS **1** AND **4** CAN TRANSFER FUEL **ONLY** INTO THEIR RESPECTIVE **INBOARD** EXT TANKS, WHICH THEN FEED THE INTERNAL FUEL TANKS. MEANING, **1** WITHOUT **2** AND **4** WITHOUT **3** ARE DEAD WEIGHT.

WP HZ											
WAYPOINTS, HAZARDS		01-50		ASSEMBLY AREA	AA		FRIENDLY AIR DEFENSE	<u>AD</u>		ENEMY AIR DEFENSE	<u>ED</u>
	COMMS CHECK POINT	CC		BATTALION	BN		FRIENDLY AIR ASSAULT	<u>AS</u>		ENEMY AIR ASSAULT	<u>ES</u>
	LANDING ZONE	LZ		BATTLE POSITION	BP		FRIENDLY AIR CAVALRY	<u>AV</u>		ENEMY AIR CAVALRY	<u>EV</u>
	PASSAGE POINT	PP		BRIDGE OR GAP	BR		FRIENDLY AIRBORNE	<u>AB</u>		ENEMY AIRBORNE	<u>EB</u>
	RELEASE POINT	RP		BRIGADE	BD		FRIENDLY ARMOR	<u>AM</u>		ENEMY ARMOR	<u>AE</u>
	START POINT	SP		CHECKPOINT	*CP		FRIENDLY AR-MORED CAVALRY	<u>CA</u>		ENEMY AR-MORED CAVALRY	<u>EC</u>
	WAYPOINT	*WP		COMPANY	CO		FRIENDLY AVIATION MAINT.	<u>MA</u>		ENEMY AVIATION MAINT.	<u>ME</u>
	TOWER OVER 1000' AGL	TO		CORPS	CR		FRIENDLY CHEMICAL	<u>CF</u>		ENEMY CHEMICAL	<u>CE</u>
	TOWER UNDER 1000' AGL	*TU		DIVISION	DI		FRIENDLY DE-CONTAMINATION	<u>DF</u>		ENEMY DE-CONTAMINATION	<u>DE</u>
	WIRES: POWER TRANSM. LINES	WL		FARP: FUEL ONLY	<u>FF</u>		FRIENDLY ENGINEERS	<u>EN</u>		ENEMY ENGINEERS	<u>EE</u>
	WIRES: PHONE AND ELECTRIC	WS		FARP: AMMO ONLY	<u>FM</u>		FRIENDLY ELECT-RONIC WARFARE	<u>FW</u>		ENEMY ELECT-RONIC WARFARE	<u>WR</u>
*CURSOR DROP DEFAULTS TO THIS SYMBOL UNDERLINED ITEMS ARE ABLE TO DISPLAY ICON TOGETHER WITH FREE TEXT				FARP: FUEL AND AMMO	<u>FC</u>		FRIENDLY FIELD ARTILLERY	<u>FL</u>		ENEMY FIELD ARTILLERY	<u>EF</u>
				FORWARD AS-SEMBLY AREA	FA		FRIENDLY FIXED WING	<u>WF</u>		ENEMY FIXED WING	<u>WE</u>
				GRND LIGHTS/ SMALL TOWN	<u>GL</u>		FRIENDLY ATTACK HELI	<u>AH</u>		ENEMY ATTACK HELI	<u>EK</u>
CM											
CONTROL MEASURES		51-99 POS 93-99		HOLDING AREA	HA		FRIENDLY GEN. ARMY HELI	<u>FG</u>		ENEMY GEN. ARMY HELI	<u>HG</u>
	AIR CONTROL POINT	AP		IDM SUBSCRIBER	<u>ID</u>		FRIENDLY HOSPITAL	<u>HO</u>		ENEMY HOSPITAL	<u>EH</u>
	AIRFIELD GENERAL	<u>AG</u>		NDB	<u>BE</u>		FRIENDLY INFANTRY	<u>FI</u>		ENEMY INFANTRY	<u>EI</u>
	AIRFIELD INSTRUMENT	<u>AI</u>		NUC/BIO/CHEM CONTAM. AREA	NB		FRIENDLY MECH. INFANTRY	<u>MI</u>		ENEMY MECH. INFANTRY	<u>EM</u>
	LIGHTED AIRPORT	<u>AL</u>		RAILHEAD: POINT	RH		FRIENDLY MEDICAL	<u>MD</u>		ENEMY MEDICAL	<u>EX</u>
	ARTILLERY FIRING POINT 1	<u>F1</u>		REGIMENT OR GROUP	GP		FRIENDLY TAC OPS CENTER	<u>TF</u>		ENEMY TAC OPS CENTER	<u>ET</u>
	ARTILLERY FIRING POINT 2	<u>F2</u>		US ARMY	US		FRIENDLY UNIT	<u>FU</u>		ENEMY UNIT	<u>EU</u>

							FCR TARGET SYMBOLS	
	TRACK	WHEEL	AIR DEF / MERGED		UNKN	HELO	FLYER	
LOAL STATIONARY 1500 – 8000 m								Priority of fire indicators: #1 NTS Next-To-Shoot (ready and not) #2 ANTS Alternate Next-To-Shoot
LOBL STATIONARY 500 – 1500 m								
LOBL MOVING 500 – 8000 m								

Time after the last scan when detected targets will turn stale:
 stationary: 30 sec
 moving: 5 sec



XPNDR APX-118

**MODES 1, 2, 3/A,
4, C, and S**

INTERCOM

INTERCOM

LEFT CONSOLE

- 1. FAT: FREE AIR TEMPERATURE GAGE
- 2. EXTERIOR / INTERIOR LIGHTING PANEL (PRIMARY, ANTI-COL, SIGNAL LIGHTS TEST)
- 3. STORES JETTISON PANEL
- 4. ENGINE START MOMENTARY TOGGLE SWITCHES
- 5. APU START/STOP
- 6. POWER LEVERS
- 7. MSTR IGN SWITCH
- 8. RTR BRK SWITCH
- 9. EMERGENCY PANEL
- 10. NVS MODE SWITCH
- 11. TAIL WHEEL LOCK
- 12. KU: KEYBOARD UNIT

FRONT PANEL AND RIGHT CONSOLE

- 13. VIDEO PANEL (ACM)
- 14. ARMAMENT PANEL (A/S, GND ORIDE)
- 15. LEFT MPD (MULTI-PURPOSE DISPLAY)
- 16. FIRE DETECTION / EXTINGUISHING PANEL (FIRE TESTS)
- 17. STANDBY COMPASS
- 18. MASTER WARNING / MASTER CAUTION
- 19. BRU: BORESIGHT RETICLE UNIT
- 20. EUFD: ENHANCED UP-FRONT DISPLAY
- 21. CMWS: AN/AAR-57 COMMON MISSILE WARNING SYSTEM
- 22. RIGHT MPD
- 23. STANDBY ATTITUDE INDICATOR
- 24. STANDBY ALTIMETER
- 25. COMMUNICATIONS PANEL
- 26. SQL: SQUELCH MOMENTARY TOGGLE SWITCHES
- 27. DTU: DATA TRANSFER UNIT
- 28. STANDBY AIRSPEED INDICATOR
- 29. GEN RST SWITCH
- 30. PARK BRAKE HANDLE

QUICKSTART

- 30 SET (HANDLE OUT)
- 7 TO BATT
- 5 APU ON
- 20 WAIT 20 SEC FOR 'APU ON'
- CLOSE THE CANOPY (LCTRL + C)
- 25 RLWR TO MAX
- 26 FLIP FORWARD ALL SQL SWITCHES
- 21 PWR ON, ARM, CMWS
- 15 [M] > [B2 ENG] AND MONITOR NG

4 NO.1 TO START AND WAIT 3-5 SEC

6 NO.1 TO IDLE (RALT + HOME)

WAIT 60 SEC UNTIL NG1 IS AT 66-67%

4 NO.2 TO START AND WAIT 3-5 SEC

6 NO.2 TO IDLE (RSHIFT + HOME)

WAIT 50 SEC UNTIL NG2 IS AT 66-67%

6 BOTH SMOOTHLY TO FLY (NUMPAD +)

WAIT UNTIL NR AND Np ARE AT 101%

5 APU OFF

PERFORM IHADSS BORESIGHTING:

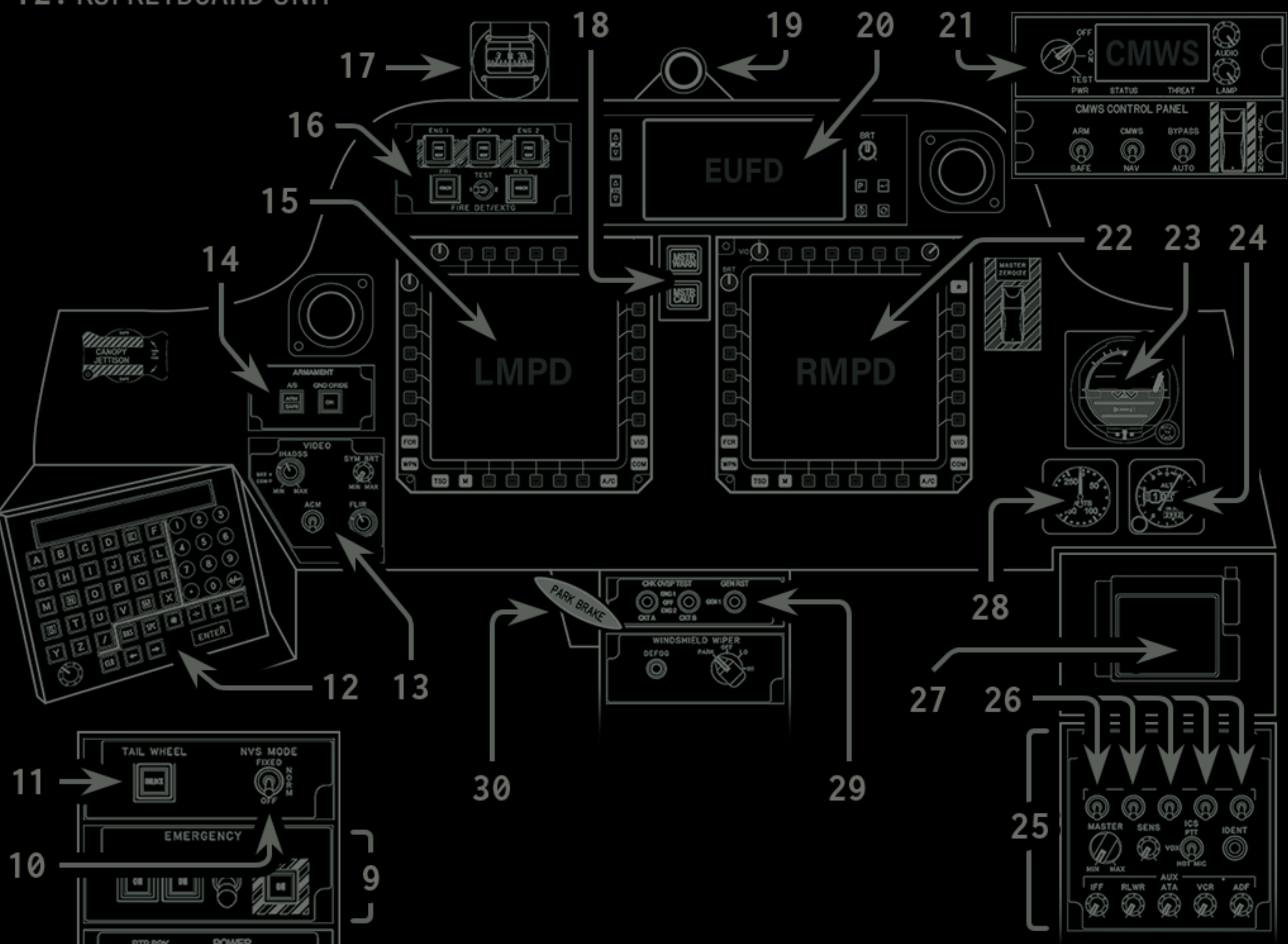
15 [WPN] > [L5 BORESIGHT] > [L4 IHADSS]

2 PRIMARY KNOB TO BRT

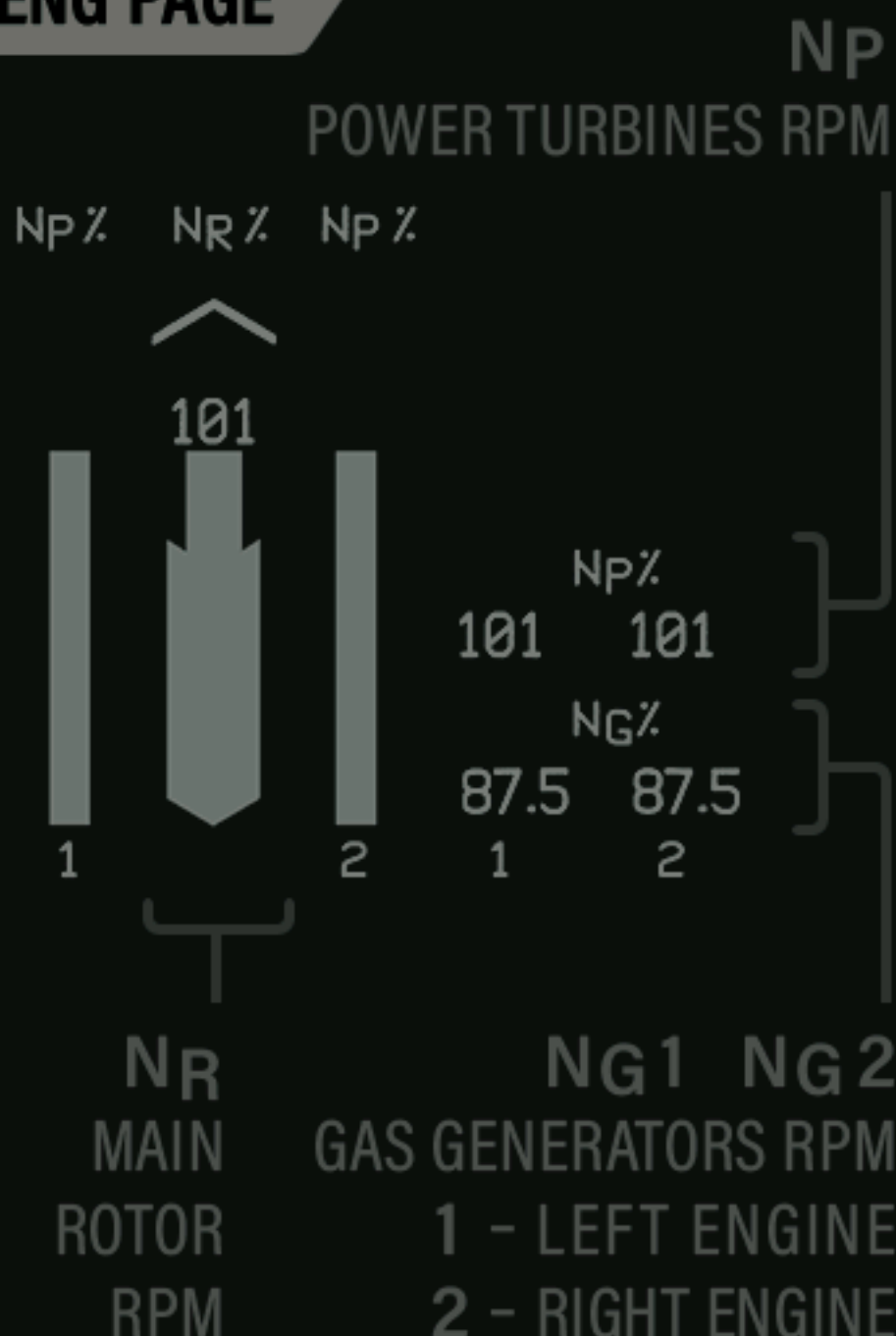
19 MOVE THE CAMERA TO ALIGN HMD RETICLE WITH BRU:



15 [L6 B/S NOW] TO CONFIRM



ENG PAGE



BEFORE TAKEOFF:

- 22 [TSD] > VERIFY THE MAP IS NOT BLACK
- 15 [FCR] > [T6 UTIL] > [R6 MMA NORM] (IF AVAILABLE)
- 30 RELEASE

AFTER TAKEOFF:

- 15 [M] > [L3 ASE] > [T1 CHAFF ARM] THEN: [T6 UTIL] > [R4 RLWR] (SOLID DOT)